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Master's Thesis

**Non-Standard Combinatorial
Ultrafilters**

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Abstract

A restricted ultrapower is a set of functions over a structure modulo an ultrafilter. These are known to preserve universal sentences true in the ground structure while potentially satisfying new sentences and can thus be applied to consistency proofs for universal theories, particularly for the theory $\forall\text{PV}$ from bounded arithmetic. This poses the question whether new consistency results can be proved when choosing an ultrafilter with additional properties. This thesis is thus dedicated to generalizing the theory of combinatorial ultrafilters over ω to non-standard models of arithmetic while restricting the combinatorial properties to objects definable over these models. We show that this restriction enables proving in ZFC the existence of such ultrafilters over various non-standard models. Finally, we demonstrate the technique of ultrapower construction by proving a slight generalization of the circuit lower bound consistency result for $\forall\text{PV}$ from [1].

Contents

1	Introduction	9
2	Combinatorial Ultrafilters	11
2.1	Basics	11
2.2	d -Ramsey Ultrafilters	15
2.3	d - P -Points and d - Q -Points	22
2.4	Strong Variants	26
3	Existence	29
3.1	Countable Ordered Structures	29
3.2	Small Ordered Structures	32
3.3	Small Parameter Sets	34
3.4	Uncountable Ordered Structures	36
3.5	Separation	43
4	Restricted Ultrapowers	45
4.1	Basics	45
4.2	Consistency of Circuit Lower Bounds	48
4.2.1	The Theory $\forall PV$	48
4.2.2	Herbrand Saturation	50
4.2.3	The Lower Bound	51
5	Conclusions	57

1 Introduction

It is a fundamental goal of complexity-theoretic research to find a logical theory that is capable of formalizing and proving the known results about algorithms and complexity and thus yields a standard notion of proof in this area. [2] Candidates for this are the theories of Bounded Arithmetic, the most prominent example of which is **PV**, introduced by Cook in [3] as a formalization of feasibly constructive proofs. It is indeed the case that “**PV** or its mild extensions seem to formalize most of contemporary complexity theory. [...] It is thus of interest to understand, given an established conjecture, whether it is provable in one of these theories or at least consistent with them.” [4] A particularly interesting class of such conjectures are circuit lower bounds. Consistency results for circuit lower bounds have for example been established in [4] and [5] by means of witnessing, in particular the KPT-theorem. The proofs are thus syntactic in nature. On the other hand, it is possible to prove such results using the model-theoretic technique of restricted ultrapowers to directly construct from a model of the given theory a second model satisfying the lower bound, as has been demonstrated in [1].

A restricted ultrapower is a set of functions over the ground model modulo an ultrafilter. The resulting structure exhibits different features and satisfies different sentences depending on the choice of the function set and the ultrafilter. This fact motivates the investigation of ultrafilters with special properties and leads to the main focus of this thesis. The goal is to generalize the theory of combinatorial ultrafilters over ω known from set theory, presented for example in [6, Chapter 11], to non-standard models of arithmetic theories. These combinatorial properties require an ultrafilter to contain special sets for a set of colorings or partitions of the universe. However, since the existence of such ultrafilters, even over ω , is not provable in **ZFC**, we will weaken the properties to talk only about definable colorings and partitions, which facilitates existence proofs for interesting classes of models, especially countable ones. From an arithmetic perspective with the goal to establish new lower bound consistencies, this restriction is acceptable because it is reasonable to assume that the colorings to which a combinatorial property will be applied in order to prove that the ultrapower satisfies a new sentence should be related to this sentence and thus definable.

Finding an application of these properties in the context of ultrapower constructions would establish an unexpected connection between set-theoretic concepts and complexity theory. In the arguably most extreme case, this could imply that additional set-theoretic axioms independent of **ZFC**, such as large cardinal axioms, have consequences for the provability of **P** versus **NP** in the theories of bounded arithmetic. Of course this is purely speculative, and such an application has yet to be found.

The thesis is divided into three main chapters. Chapter 2 develops the theory of different combinatorial ultrafilter properties in the non-standard setting. The question

1 Introduction

which ultrafilters exist for which models is discussed in Chapter 3. Finally, Chapter 4 proves a slight generalization of the main theorem from [1] in order to demonstrate how consistency results can be proved using an ultrapower construction.

2 Combinatorial Ultrafilters

This chapter pursues the goal of developing a theory of combinatorial ultrafilters in the non-standard, definable setting. It starts with a section on basic definitions and motivation in the context of the standard theory over ω , then proceeds with the analysis of the central notion of d -Ramsey ultrafilters, followed by the closely related d - P -points and d - Q -points, and finishes with a section investigating natural strengthenings of these properties.

2.1 Basics

The entire theory of combinatorial ultrafilters builds upon Ramsey's famous result in combinatorics showing the existence of infinite monochromatic subsets of ω for all colorings of the n -element subsets of ω with finitely many colors.

Definition 2.1.1. For an infinite set S and a cardinal $\kappa \leq |S|$, let $[S]^\kappa$ denote the set of all subsets of S of size κ . A function $\pi : [S]^n \rightarrow r$ with $n, r \in \omega$ is called n -ary coloring. An infinite set $H \subseteq S$ is *homogeneous* for a coloring $\pi : [S]^n \rightarrow r$ if $\pi \upharpoonright [H]^n$, i. e., the restriction of π to $[H]^n$, is constant.

The following proof is taken from [7, Chapter 9].

Theorem 2.1.2 (Ramsey). *For any number $n \in \omega$, for any positive number $r \in \omega$, for any $S \in [\omega]^\omega$, and for any coloring $\pi : [S]^n \rightarrow r$, there is always an $H \in [S]^\omega$ such that H is homogeneous for π , i. e., the set $[H]^n$ is monochromatic. [6, Chapter 4]*

Proof. Without loss of generality assume $S = \omega$. Proceed by induction over n . For $n = 1$, the statement is trivial, so assume that it is true for n and conclude that it holds for $n + 1$. Let $f : [\omega]^{n+1} \rightarrow r$ be an arbitrary $n + 1$ -ary coloring. For each $a \in \omega$, define the function $f_a : (\omega \setminus \{a\})^n \rightarrow r$ such that $f_a(X) = f(X \cup \{a\})$ for each $X \in (\omega \setminus \{a\})^n$. Then, by the induction hypothesis, we find for each $a \in \omega$ and each infinite subset S of ω an infinite homogeneous set $H_a^S \subseteq S \setminus \{a\}$ for the coloring f_a . We inductively construct an infinite set $A := \{a_i \mid i \in \omega\}$ as follows. Set $S_0 := \omega, a_0 := 0$, and for each $i \in \omega$ let $S_{i+1} := H_{a_i}^{S_i}$ and $a_{i+1} := \bigcap S_{i+1} \setminus (a_i + 1)$. Then, for each $i \in \omega$ we have that f_{a_i} is constant on $[A \setminus (a_i + 1)]^n$. Define the unary coloring $g : A \rightarrow r$ of A mapping a_i to this constant value of f_{a_i} . Now, take an infinite homogeneous set $H \subseteq A$ for the coloring g and let c be its constant value. Then, H is also homogeneous for f because for $b_1 < \dots < b_{n+1}$ in H it holds that $f(\{b_1, \dots, b_{n+1}\}) = f_{b_1}(\{b_2, \dots, b_{n+1}\}) = g(b_1) = c$. $\square$

Having shown the existence of infinite homogeneous sets for all colorings, we are interested in the question whether one can collect such sets in an ultrafilter, leading to the definition of Ramsey ultrafilters.

Definition 2.1.3. Let S be an infinite set. A set $\mathcal{U} \subseteq [S]^{\geq \omega}$ is called *filter* if

- $\emptyset \notin \mathcal{U}$,
- $S \in \mathcal{U}$,
- if $X \subseteq Y, X \in \mathcal{U}$, then $Y \in \mathcal{U}$,
- if $X_1, \dots, X_n \in \mathcal{U}$, then $\bigcap_{i \in \{1, \dots, n\}} X_i \in \mathcal{U}$.

It is called *ultrafilter* if it is a maximal filter. Equivalently, it is maximal if for each $X \subseteq S$ either $X \in \mathcal{U}$ or $S \setminus X \in \mathcal{U}$.

Definition 2.1.4. An ultrafilter $\mathcal{U}$ over ω is called *Ramsey* if for each binary coloring $\pi : [\omega]^2 \rightarrow 2$ there exists an infinite homogeneous set $H \in \mathcal{U}$.

It is important to note that this property implies the seemingly stronger inclusion of infinite homogeneous sets for all n -ary colorings $\pi : [\omega]^n \rightarrow r$ with $n, r \in \omega$. While this combinatorial property is interesting in itself, it cannot be unconditionally applied in the context of a model-theoretic construction, since the existence of Ramsey ultrafilters is in fact independent of ZFC.

Theorem 2.1.5. *The existence of Ramsey ultrafilters over ω is independent of ZFC. [7]*

Based on a proof in [7], we show that the existence of Ramsey ultrafilters follows from the continuum hypothesis and, since CH is well-known to be consistent with ZFC, this implies the consistency of Ramsey ultrafilters. Establishing the consistency of their non-existence can be achieved using Mathias forcing. For a full proof see for example [6, Chapter 26].

Theorem 2.1.6. *If $2^{\aleph_0} = \aleph_1$, then a Ramsey ultrafilter exists. [7, Chapter 7]*

Proof. Let π_α for $\alpha \in \aleph_1$ enumerate all binary colorings of ω . We inductively construct a family with the strong finite intersection property as follows. Choose for X_0 any infinite homogeneous set of the coloring π_0 . For a successor ordinal $\alpha + 1$ assume that X_α is defined as an infinite set and define $X_{\alpha+1} \subseteq X_\alpha$ to be an infinite homogeneous set of the coloring $\pi_{\alpha+1} \upharpoonright [X_\alpha]^2$. For a limit ordinal λ , assume that all X_α with $\alpha < \lambda$ are defined. By the continuum hypothesis, λ is countable, thus it has cofinality $\aleph_0$. In particular, there exists a set $I \subseteq \lambda$ that is unbounded in λ and has order type ω . It follows that there exists an infinite set Y_λ such that $Y_\lambda \setminus X_\alpha$ for all $\alpha \in I$ is finite. By the unboundedness of I , $Y_\lambda \setminus X_\alpha$ is finite for all $\alpha \in \lambda$. Then, $X_\lambda \subseteq Y_\lambda$ can be defined as an infinite homogeneous set of the coloring $\pi_\lambda \upharpoonright [Y_\lambda]^2$. Clearly, the family $\{X_\alpha \mid \alpha \in \aleph_1\}$ has the strong finite intersection property, implying that it can be extended to an ultrafilter, which is Ramsey by construction. $\square$

The independence result in Theorem 2.1.5 makes Ramsey ultrafilters unsuitable for unconditional model-theoretic constructions, which will lead to the weaker notion of d -Ramsey ultrafilters introduced below. Further, we are interested in combinatorial ultrafilters over non-standard orders, shifting the focus from ω to the following class of structures. Fix a countable language L containing the binary relation symbol $<$ and an L -structure $\mathfrak{A}$, where A denotes its universe, such that $<^{\mathfrak{A}}$ is a right-discrete linear order with a minimum and no maximum, meaning that $\mathfrak{A}$ satisfies the following theory.

Definition 2.1.7. The $\{<\}$ -theory $T_{<}$ consists of the following axioms:

$\forall x \, x \not< x$	(irreflexivity)
$\forall x \forall y (x < y \rightarrow y \not< x)$	(asymmetry)
$\forall x \forall y \forall z ((x < y \wedge y < z) \rightarrow x < z)$	(transitivity)
$\forall x \forall y (x \neq y \rightarrow (x < y \vee y < x))$	(linearity)
$\forall x \exists y \forall z (x < y \wedge (z < y \rightarrow z \leq x))$	(right-discreteness)
$\exists x \forall y \, x \leq y$	(minimum)
$\forall x \exists y \, x < y$	(no maximum)

Here, we write $x \not< y$ for $\neg x < y$ and $x \leq y$ for $x < y \vee x = y$.

This, in particular, comprises all transfinite ordinals and models of arithmetic. We will refer to an L -structure modeling the theory $T_{<}$ as an *ordered structure*. This constitutes a slight abuse of common terminology since, first, a particular kind of order is required and, second, the language L is allowed to contain countably many symbols aside from $<$. This choice, however, is motivated by the fact that the vast majority of properties investigated in this thesis are properties of the order $<^{\mathfrak{A}}$. Yet, we cannot dispense with the larger language L , because we are explicitly interested in models of bounded arithmetic, which interpret the countable language **PV**.

Central to this non-standard theory of combinatorial ultrafilters is the generalization of finiteness to boundedness. In particular, we generalize free ultrafilters over ω , i. e., ultrafilters containing only infinite sets, to unbounded ultrafilters.

Definition 2.1.8. Define $[a] := \{b \in A \mid b < a\}$. A set $X \subseteq A$ is *bounded* if there exists an $a \in A$ such that $X \subseteq [a]$. Otherwise it is *unbounded*. An ultrafilter $\mathcal{U}$ is *unbounded* if every set in $\mathcal{U}$ is unbounded.

In the following, we tacitly assume every ultrafilter to be unbounded. The following property is the essential notion for showing that a family of unbounded sets is contained in an unbounded ultrafilter.

Definition 2.1.9. A family $\mathcal{X}$ of unbounded subsets of A has the *finite intersection property* if for all finite subsets $\mathcal{Y} \subseteq \mathcal{X}$ the intersection $\bigcap \mathcal{Y}$ is non-empty. It has the *unbounded finite intersection property* if for all finite subsets $\mathcal{Y} \subseteq \mathcal{X}$ the intersection $\bigcap \mathcal{Y}$ is unbounded.

Proposition 2.1.10. *Every family $\mathcal{X}$ of unbounded sets with the unbounded finite intersection property can be extended to an unbounded ultrafilter.*

Proof. By the ultrafilter theorem it suffices to show that $\mathcal{X} \cup \{A \setminus [a] \mid a \in A\}$ has the finite intersection property. So let $X_1, \dots, X_n \in \mathcal{X}, A \setminus [a_1], \dots, A \setminus [a_m]$ with $a_1 < \dots < a_m$ from A be a finite subfamily. Then, $X := \bigcap_{i \in \{1, \dots, n\}} X_i$ is unbounded by assumption, $\bigcap_{i \in \{1, \dots, m\}} A \setminus [a_i] = A \setminus [a_m]$ is co-bounded, and thus the intersection $(\bigcap_{i \in \{1, \dots, n\}} X_i) \cap (\bigcap_{i \in \{1, \dots, m\}} A \setminus [a_i]) = X \setminus [a_m]$ is clearly unbounded as well. $\square$

This generalization to unbounded sets raises the question how large unbounded sets are at least. This motivates the investigation of cofinality, a concept that will occur throughout the thesis.

Definition 2.1.11. The *cofinality* of $\mathfrak{A}$, denoted by $\text{cf}(\mathfrak{A})$, is the smallest cardinal κ such that there exists an unbounded set of size κ .

Lemma 2.1.12. *$\text{cf}(\mathfrak{A})$ is the smallest cardinal κ such that there exists an unbounded, increasing $s : \kappa \rightarrow A$. In particular, $\text{cf}(\mathfrak{A})$ is regular. [8]*

Proof. It suffices to find an unbounded, increasing $s : \kappa \rightarrow A$. Let $\{a_\alpha \mid \alpha \in \text{cf}(\mathfrak{A})\}$ be unbounded and assume that $s(\alpha)$ is defined for all $\alpha < \beta$ with $\beta \in \text{cf}(\mathfrak{A})$. Both $\{a_\alpha \mid \alpha < \beta\}$ and $\{s(\alpha) \mid \alpha < \beta\}$ are bounded. Choose for $s(\beta)$ a common bound of those sets. Then, s is unbounded and increasing by construction. $\square$

Definition 2.1.13. A set $S \subseteq A$ is called *cofinal* if there exists an unbounded, increasing function $s : \text{cf}(\mathfrak{A}) \rightarrow A$ with $\text{im}(s) = S$, where $\text{im}(s)$ denotes the set $\{s(\alpha) \mid \alpha \in \text{cf}(\mathfrak{A})\}$.

Finally, as mentioned before, we will focus on weak versions of combinatorial ultrafilter properties from the standard theory due to the independence result in Theorem 2.1.5. This weakening is done by restricting the universal quantifiers in the definitions to the definable domain. In the case of Ramsey ultrafilters this means requiring the inclusion of homogeneous sets only for definable colorings. We will see in Chapter 3 that ultrafilters with such definable versions of combinatorial properties provably exist for certain classes of ordered structures.

Definition 2.1.14. Let $L(B)$ for $B \subseteq A$ denote the language L expanded by new constant symbols c_b for each $b \in B$ and let $\mathfrak{A}_B$ denote the expansion of $\mathfrak{A}$ to $L(B)$ by interpreting each c_b by b . A subset $X \subseteq A$ is *definable over $B \subseteq A$* if there exists an $L(B)$ -formula $\varphi(x)$ such that $X = \varphi(\mathfrak{A}) := \{a \in A \mid \mathfrak{A}_B \models \varphi(a)\}$. It is *definable* if it is definable over A .

Definition 2.1.15. A coloring $\pi : [A]^n \rightarrow r$ with $n, r \in \omega$ is called *definable* if there exists a sequence of formulas $\varphi_0(x_1, \dots, x_n), \dots, \varphi_{r-1}(x_1, \dots, x_n)$ such that $\mathfrak{A} \models \varphi_c(a_1, \dots, a_n)$ if and only if $\pi(\{a_1, \dots, a_n\}) = c$ for all $\{a_1, \dots, a_n\} \in [A]^n, c \in r$.

For $r = 2$, we may say that a coloring $\pi : [A]^n \rightarrow 2$ is defined by a single formula $\varphi(x_1, \dots, x_n)$ and mean that it is defined by the formulas

$$\begin{aligned} \varphi_0(x_1, \dots, x_n) &:= \neg \varphi(x_1, \dots, x_n), \\ \varphi_1(x_1, \dots, x_n) &:= \varphi(x_1, \dots, x_n). \end{aligned}$$

2.2 d -Ramsey Ultrafilters

This section focusses on the analysis of d -Ramsey ultrafilters and loosely follows the chapter on happy families found in [6, Chapter 11], including definable variants of selectivity and the notion of diagonalization, which are introduced below. Fix a countable language L and an ordered L -structure $\mathfrak{A}$ modeling $T_{<}$.

Definition 2.2.1. An ultrafilter $\mathcal{U}$ is d^n -Ramsey for $n \in \omega$ if for every definable n -ary coloring $\pi : [A]^n \rightarrow 2$ there exists an unbounded homogeneous set $X \in \mathcal{U}$. For $n = 2$, we may omit the superscript and simply write d -Ramsey. It is $d^{<\omega}$ -Ramsey if it is d^n -Ramsey for all $n \in \omega$.

While, contrary to Ramsey ultrafilters over ω , it is unknown whether d^2 -Ramsey and $d^{<\omega}$ -Ramsey ultrafilters are equivalent and, hence, this terminological distinction has to be made, the former capture exactly the property that is required for the coming characterization results, motivating the choice to simply call them d -Ramsey. See Section 2.4 for a brief discussion on why the equivalence of d -Ramsey and $d^{<\omega}$ -Ramsey ultrafilters might fail in general. On the other hand, it can be shown that the restriction to only two colors is without loss of generality.

Remark 2.2.2. If an ultrafilter $\mathcal{U}$ is d^n -Ramsey, then it contains unbounded homogeneous sets for every definable coloring $\pi : [A]^n \rightarrow r$ with $r \in \omega$.

Proof. Prove this claim by induction over r . It trivially holds for $r = 1$ and by assumption also for $r = 2$. Now suppose it holds for a fixed $r \geq 2$ and consider an arbitrary coloring $\pi : [A]^n \rightarrow r \cup \{r\}$ definable by formulas $\varphi_0(x_1, \dots, x_n), \dots, \varphi_r(x_1, \dots, x_n)$. Define the colorings

$$\rho_0(\{a_1, \dots, a_n\}) := \begin{cases} 1 & \pi(\{a_1, \dots, a_n\}) \in \{0, 1\} \\ \pi(\{a_1, \dots, a_n\}) & \text{otherwise} \end{cases}$$

and

$$\rho_1(\{a_1, \dots, a_n\}) := \begin{cases} 1 & \pi(\{a_1, \dots, a_n\}) \in \{1, 2\} \\ \pi(\{a_1, \dots, a_n\}) & \text{otherwise} \end{cases}.$$

These colorings are clearly definable and use only r colors, hence, $\mathcal{U}$ contains homogeneous sets X_0 and X_1 for each of them. If the constant value of ρ_0 on X_0 is not 1, then X_0 is a homogeneous set for π . Analogously, if the constant value of ρ_1 on X_1 is not 1, then X_1 is a homogeneous set for π . However, if both these values equal 1, then the set $X := X_0 \cap X_1 \in \mathcal{U}$ is a homogeneous set. To see this, take $B \in [X]^n$ and observe that $\pi(B) \in \{0, 1\}$ by $\rho_0(B) = 1$ and $\pi(B) \in \{1, 2\}$ by $\rho_1(B) = 1$, giving $\pi(B) = 1$. $\square$

Further, the following remark shows that d^n -Ramsey and d^m -Ramsey properties for distinct n, m are not independent but form a hierarchy of properties of non-decreasing strength with increasing arity. Having observed this relation, proving the equivalence of d -Ramsey and $d^{<\omega}$ -Ramsey ultrafilters reduces to showing the other direction, i. e., that this hierarchy collapses.

Remark 2.2.3. *If an ultrafilter $\mathcal{U}$ is d^n -Ramsey, then it is d^m -Ramsey for all $1 \leq m \leq n$.*

Proof. Prove this by finite induction over m from n to 1. Assuming it holds for some $2 \leq m \leq n$, take an arbitrary definable $(m-1)$ -ary coloring $\pi : [A]^{m-1} \rightarrow 2$. Now construct the definable m -ary coloring $\hat{\pi} : [A]^m \rightarrow 2$ with $\hat{\pi}(B) := \pi(B \setminus \{\max B\})$ for $B \in [A]^m$. By the induction hypothesis, $\mathcal{U}$ contains a homogeneous set X for $\hat{\pi}$. Denote the constant value by c . This set is also homogeneous for π because $\pi(B) = \hat{\pi}(B \cup \{b\}) = c$ for $B \in [X]^{m-1}$ and $b \in X$ with $b > \max B$. Furthermore, one can find such a b since X is unbounded. $\square$

As proved in [6, Chapter 11], an ultrafilter over ω is Ramsey if and only if it contains homogeneous sets for all partitions of ω . Moving to the non-standard, definable setting, it can be shown that d -Ramsey ultrafilters have a related, weaker property restricted to definable partitions.

Definition 2.2.4. A partition $\mathcal{P}$ of A is *represented* by a function $\pi : A \rightarrow A$ if for every $P \in \mathcal{P}$ there exists an $a \in A$ such that $\pi^{-1}(a) = P$. A function is *definable* if its graph is definable. A partition is *definable* if it is represented by a definable function.

We will often write $\mathcal{P}_f$ for the partition represented by a given function $f : A \rightarrow A$ and, conversely, $f_{\mathcal{P}}$ for a function representing a given partition $\mathcal{P}$.

Definition 2.2.5. An ultrafilter $\mathcal{U}$ is *d -selective* if for every definable partition $\mathcal{P}$ of A either $\mathcal{U} \cap \mathcal{P} \neq \emptyset$ or there exists an $X \in \mathcal{U}$ such that $|X \cap P| \leq 1$ for $P \in \mathcal{P}$.

Lemma 2.2.6. *If an ultrafilter $\mathcal{U}$ is d -Ramsey, then it is d -selective.*

Proof. Let $\varphi(x, y)$ define $\mathcal{P}$. Consider the coloring $\pi : [A]^2 \rightarrow 2$ defined by

$$\psi(x, y) := \exists z \exists z' (z \neq z' \wedge \varphi(x, z) \wedge \varphi(y, z')),$$

intuitively stating that elements x and y lie in different pieces of $\mathcal{P}$. Choose $X \in \mathcal{U}$ such that $\pi \upharpoonright [X]^2$ is constant. Then, X has the desired properties because all elements in X are either contained in the same piece of $\mathcal{P}$, verifying the first case, or in different pieces, verifying the second case. $\square$

Observing that partitions of A and functions on A are conceptually the same by Definition 2.2.4, an equivalent formulation of d -selectivity is to say that an ultrafilter contains for each function a set on which it is constant or injective. This indirectly relates the d -Ramsey property to functions, but the following result shows that d -Ramsey ultrafilters give something slightly stronger.

Proposition 2.2.7. *Let $\mathcal{U}$ be a d -Ramsey ultrafilter over $\mathfrak{A}$. Then, for every definable function $f : A \rightarrow A$, there exists a set $X \in \mathcal{U}$ such that $f \upharpoonright X$ is increasing, decreasing, or constant.*

Proof. Let $\varphi(x, y)$ define f . Consider the coloring $\pi_0 : [A]^2 \rightarrow 2$ defined by the formula

$$\psi_0(x, y) := \exists z(\varphi(x, z) \wedge \varphi(y, z)),$$

which intuitively states that $f(x) = f(y)$. Then, $\mathcal{U}$ contains an unbounded homogeneous set X_0 for this coloring, for which it holds that $f \upharpoonright X_0$ is either constant or injective. If it is constant, we are done, so assume it is injective. Then, consider the coloring $\pi_1 : [A]^2 \rightarrow 2$ defined by the formula

$$\psi_1(x, y) := \psi'(x, y) \vee \psi'(y, x)$$

with

$$\psi'(x, y) := \exists z_0 \exists z_1 (\varphi(x, z_0) \wedge \varphi(y, z_1) \wedge x \leq y \wedge z_0 \leq z_1),$$

where the latter intuitively states that $x \leq y$ and $f(x) \leq f(y)$. Then, there exists a homogeneous set $X_1 \in \mathcal{U}$, hence, $f \upharpoonright X_1$ is either decreasing or non-decreasing. If it is decreasing, we are done, so assume it is non-decreasing. Then, $f \upharpoonright X$ for $X := X_0 \cap X_1 \in \mathcal{U}$ is both non-decreasing and injective and thus increasing. $\square$

Next, the notion of cofinal ultrafilters is introduced and put into relation to d -selective ultrafilters. This property is crucial for several of the following proofs because it in some cases allows to ignore the order $<^{\mathfrak{A}}$ and concentrate on the regular cardinal $\text{cf}(\mathfrak{A})$.

Definition 2.2.8. An ultrafilter $\mathcal{U}$ is *cofinal* if it contains a cofinal subset of A .

Lemma 2.2.9. *If there exists a definable cofinal set, then every d -selective ultrafilter is cofinal.*

Proof. Let $\mathcal{U}$ be d -selective and $s : \text{cf}(\mathfrak{A}) \rightarrow A$ unbounded and increasing with the set $\text{im}(s)$ defined by some formula $\sigma(x)$. Let $\mathcal{P}$ be the partition defined by

$$\varphi(x, y) := \sigma(y) \wedge x \leq y \wedge \forall z((z < y \wedge \sigma(z)) \rightarrow z < x),$$

stating that x is contained in the piece $\varphi(\mathfrak{A}, y)$ if y is the smallest element of $\text{im}(s)$ greater or equal to x . By Lemma 2.2.6, there exists an $X \in \mathcal{U}$ such that $|X \cap P| \leq 1$ for all $P \in \mathcal{P}$. This is true by the unboundedness of $\mathcal{U}$ and the fact that every piece of $\mathcal{P}$ is bounded, which implies $\mathcal{U} \cap \mathcal{P} = \emptyset$. Then, X is well-ordered of order type at most $\text{cf}(\mathfrak{A})$. Since it is unbounded, it is of type $\text{cf}(\mathfrak{A})$. $\square$

Note that, because the notion of well-foundedness is not expressible in first-order logic, the assumption that an unbounded set of order type $\text{cf}(\mathfrak{A})$ can be defined is difficult to justify in general. However, since we allow arbitrary countable languages, it is trivially true in ordered structures interpreting some unary relation symbol in the required way and of course in the standard model of arithmetic $\mathbb{N}$.

Beyond that, one can dispense with the condition that there exists some definable cofinal set by strengthening d -selectivity in a self-evident way, namely by simply assuming that making an appropriate cofinal set definable does not violate d -selectivity.

Definition 2.2.10. An ultrafilter $\mathcal{U}$ is *robustly d -selective* if there exists a cofinal subset S of $\mathfrak{A}$ such that $\mathcal{U}$ is d -selective in the $L \cup \{C\}$ -structure $(\mathfrak{A}, S)$ for a new unary relation symbol C . Robust variants of related ultrafilter properties, such as d -Ramsey, are defined analogously.

The analogous result of Lemma 2.2.9 now does not depend anymore on the existence of a definable cofinal set, because one can consider the robustly d -Ramsey ultrafilter in the expansion $(\mathfrak{A}, S)$, where Lemma 2.2.9 can directly be applied. This works because the cofinality property of ultrafilters is obviously preserved under taking reducts.

Corollary 2.2.11. *Every robustly d -selective ultrafilter is cofinal.*

Proof. Let $\mathcal{U}$ be robustly d -selective. Then, there exists a cofinal set S such that $\mathcal{U}$ is a d -selective ultrafilter of $(\mathfrak{A}, S)$. By Lemma 2.2.9, $\mathcal{U}$ is cofinal. $\square$

When it comes to the question of existence, it is helpful to investigate what the set of robustly d -Ramsey ultrafilters over $\mathfrak{A}$ looks like. In fact, it comprises exactly the d -Ramsey ultrafilters of the expansions $(\mathfrak{A}, S)$ for all cofinal $S \subseteq A$.

Remark 2.2.12. *Let $\mathcal{S}$ be the set of cofinal subsets of $\mathfrak{A}$. The set of robustly d -Ramsey ultrafilters is exactly the set $\bigcup_{S \in \mathcal{S}} \{\mathcal{U} \mid \mathcal{U} \text{ is a } d\text{-Ramsey ultrafilter of } (\mathfrak{A}, S)\}$.*

Proof. The inclusion from left to right is clear by definition. The inclusion from right to left follows from the observation that the d -Ramsey property is preserved under taking reducts. This is obvious since reducing the language can only reduce the set of definable colorings that have to be accommodated by a d -Ramsey ultrafilter. $\square$

As we will see in Chapter 3, there are classes of ordered structures closed under countable expansions for which d -Ramsey ultrafilters exist, including the class of countable ordered structures. Hence, the assumption that robustly d -Ramsey ultrafilters exist can be sufficiently motivated contrary to assuming the existence of a definable cofinal set. Nevertheless, the following results use the latter assumption for the sake of simplicity. In each case, an argument as in the proof of Corollary 2.2.11 can be made since all of the following ultrafilter properties are preserved under taking reducts.

Proceeding with the characterization of d -Ramsey ultrafilters, we investigate the notion of diagonalization. In [6, Chapter 11], this property is used to show that selective ultrafilters over ω are Ramsey. As we will see later, however, difficulties arise when trying to pursue a similar argument in the definable, non-standard setting.

Definition 2.2.13. Let $(X_a)_{a \in A}$ be a non-increasing chain of subsets of A . A set X *diagonalizes* $(X_a)_{a \in A}$ if $X \setminus [a+1] \subseteq X_a$ for all $a \in X$. An ultrafilter $\mathcal{U}$ is *diagonalizing* if it contains a diagonalizing set for every non-increasing chain in $\mathcal{U}$. It is *d -diagonalizing* if it contains a diagonalizing set for every definable non-increasing chain in $\mathcal{U}$. A family $(X_a)_{a \in A}$ is *definable* if there exists a formula $\varphi(x, y)$ such that $X_a = \varphi(a, \mathfrak{A})$ for all $a \in A$.

The proof below is based on a proof given in [8] for an analogous result in the general setting omitting the condition of definability.

Proposition 2.2.14. *Ultrafilters that are d -selective and contain a definable cofinal set are d -diagonalizing.*

Proof. Let $\mathcal{U}$ be a d -selective ultrafilter, let $s : \text{cf}(\mathfrak{A}) \rightarrow A$ be unbounded and increasing with $\text{im}(s) \in \mathcal{U}$ defined by some formula $\sigma(x)$, and let $(X_a)_{a \in A}$ be a non-increasing chain in $\mathcal{U}$ defined by $\chi(x, y)$. If $Z := \bigcap_{a \in A} X_a \in \mathcal{U}$, then Z is a diagonalizing set, so assume $Z \notin \mathcal{U}$. Z is definable by $\zeta(y) := \forall x \chi(x, y)$. Note that $Z = \bigcap_{a \in \text{im}(s)} X_a$ since $(X_a)_{a \in A}$ is non-increasing and $\text{im}(s)$ is unbounded. Let

$$\begin{aligned} \pi(x, y) := & y = s(0) \wedge (\neg \chi(s(0), x) \vee \zeta(x)) \\ & \vee y \neq s(0) \wedge \sigma(y) \wedge \neg \chi(y, x) \wedge \forall z ((\sigma(z) \wedge z < y) \rightarrow \chi(z, x)) \end{aligned}$$

define the partition $\mathcal{P} := \{Y_{s(\alpha)} \mid \alpha \in \text{cf}(\mathfrak{A})\}$ with $Y_0 = A \setminus X_0 \cup Z$ and $Y_{s(\alpha)} = (\bigcap_{b < s(\alpha)} X_b) \setminus X_{s(\alpha)}$ for all positive $\alpha \in \text{cf}(\mathfrak{A})$ and note that $Y_0 \notin \mathcal{U}$. Then, by the d -selectivity of $\mathcal{U}$, there is a $Y \in \mathcal{U}$ such that $|Y \cap Y_{s(\alpha)}| \leq 1$ for all $\alpha \in \text{cf}(\mathfrak{A})$

Note that the set $Y \cap \bigcup_{\beta \leq \alpha} Y_\beta$ is bounded, say by b_α . Now, inductively define an increasing function $g : \text{cf}(\mathfrak{A}) \rightarrow \text{cf}(\mathfrak{A})$. Define $g(0) > 0$ and for $\alpha \in \text{cf}(\mathfrak{A})$ assume that $g(\beta)$ is defined for all $\beta < \alpha$ and choose for $g(\alpha)$ the minimal value in $\text{cf}(\mathfrak{A})$ larger than all $g(\beta)$ such that $b_\alpha < s(g(\alpha))$. This value exists by the regularity of $\text{cf}(\mathfrak{A})$. Then, g is increasing and has the property that

$$Y \setminus [s(g(\alpha))] \subseteq X_{s(\alpha)} \quad (2.1)$$

for all $\alpha \in \text{cf}(\mathfrak{A})$: For $a \in Y \setminus [s(g(\alpha))]$, there exists a unique β with $a \in Y_{s(\beta)}$. By $b_\alpha < s(g(\alpha)) \leq a$, it follows that $a \notin \bigcup_{\gamma < \alpha} Y_{s(\gamma)}$, which implies that $\alpha < \beta$ and thus $Y_{s(\beta)} \subseteq X_{s(\alpha)}$, yielding $a \in X_{s(\alpha)}$.

Now inductively define an increasing function $h : \text{cf}(\mathfrak{A}) \rightarrow \text{cf}(\mathfrak{A})$ by setting $h(0) := 0$, $h(\alpha + 1) := g(h(\alpha))$, and $h(\lambda) := \sup_{\alpha < \lambda} h(\alpha)$ for a limit ordinal λ . This supremum exists again by the regularity of $\text{cf}(\mathfrak{A})$. Then, h is increasing since g is increasing and $g(0) > 0$, and it has the property that

$$\text{if } s(\beta) < s(h(\alpha + 1)), \text{ then } s(g(\beta)) < s(h(\alpha + 2)) \quad (2.2)$$

for all $\beta \in \text{cf}(\mathfrak{A})$. This follows from the fact that both s and g are increasing because $s(\beta) < s(h(\alpha + 1))$ implies $\beta < h(\alpha + 1)$, which in turn implies $g(\beta) < h(\alpha + 2)$, finally yielding $s(g(\beta)) < s(h(\alpha + 2))$.

Now partition A into three pieces Z_0, Z_1, Z_2 where

- Z_0 is the union of intervals $[s(h(\alpha)), s(h(\alpha + 1)))$ for even $\alpha \in \text{cf}(\mathfrak{A})$,
- Z_1 is the union of intervals $[s(h(\alpha)), s(h(\alpha + 1)))$ for odd $\alpha \in \text{cf}(\mathfrak{A})$,
- Z_2 is the union of the sets $\bigcap_{\alpha < \lambda} [s(h(\alpha)), s(h(\alpha + 1)))$ for limit ordinals $\lambda \in \text{cf}(\mathfrak{A})$.

Since $\mathcal{U}$ is an ultrafilter, it contains exactly one of these sets, but it does not contain Z_2 because $Z_2 \cap \text{im}(s) = \emptyset$. We now distinguish the two cases that either $Z_0 \in \mathcal{U}$ or $Z_1 \in \mathcal{U}$.

2 Combinatorial Ultrafilters

Assume that $Z_0 \in \mathcal{U}$, giving $Z_0 \cap Y \in \mathcal{U}$. By d -selectivity, $\mathcal{U}$ contains a set

$$Z := \{c_\alpha \mid \alpha \in \text{cf}(\mathfrak{A}) \text{ even}\} \subseteq \text{im}(s)$$

such that

$$c_\alpha \in [s(h(\alpha)), s(h(\alpha + 1))) \quad (2.3)$$

for each even $\alpha \in \text{cf}(\mathfrak{A})$. Then, the set $Y \cap Z \in \mathcal{U}$ diagonalizes the family $(X_a)_{a \in A}$: Let $\alpha \in \text{cf}(\mathfrak{A})$ be even such that $c_\alpha \in Y \cap Z$ and consider the unique β with $c_\alpha = s(\beta)$. In order to verify diagonalization, it remains to show that

$$(Y \cap Z) \setminus [s(\beta) + 1] \subseteq X_{s(\beta)}.$$

An element of the left-hand side equals c_γ for some even $\gamma \in \text{cf}(\mathfrak{A})$ with $\gamma > \alpha$ and consequently $\gamma \geq \alpha + 2$, therefore

$$s(h(\alpha + 2)) \leq s(h(\gamma)) \leq c_\gamma.$$

The first inequality follows because s is increasing, while the second inequality is a consequence of (2.3). Again by (2.3), it follows that $s(\beta) = c_\alpha < s(h(\alpha + 1))$, and thus $s(g(\beta)) < s(h(\alpha + 2))$ by (2.2). This gives $s(g(\beta)) < c_\gamma$ and therefore $c_\gamma \in Y \setminus [s(g(\beta))]$. Finally, (2.1) yields $c_\gamma \in X_{s(\beta)}$.

The second case that $Z_1 \in \mathcal{U}$ is analogous to the first case and simply talks about odd $\alpha \in \text{cf}(\mathfrak{A})$ instead even ones. $\square$

As mentioned earlier, one essential aspect of the non-standard theory of combinatorial ultrafilters is the generalization of finiteness to boundedness. However, a lot of arguments in the standard theory take advantage of exactly the fact that these notions coincide and, hence, ultrafilters are by definition closed under families of sets with bounded index set. In the non-standard theory, some proofs may require a stronger closure property than ω -closedness.

Definition 2.2.15. Let κ be a cardinal. An ultrafilter $\mathcal{U}$ is κ -closed if it contains an $X \subseteq \bigcap_{\alpha < \lambda} X_\alpha$ for every $(X_\alpha)_{\alpha < \lambda}$ in $\mathcal{U}$ with $\lambda < \kappa$.

The next two propositions use d -diagonalization to prove the d -Ramsey property and d -selectivity, respectively, under rather strong additional assumptions. At the end of Section 2.4, it is discussed why a general equivalence of the d -Ramsey, d -selectivity, and d -diagonalization properties, which one could assume based on the standard theory, seems unlikely.

Proposition 2.2.16. *If there exists a definable cofinal set, then $\text{cf}(\mathfrak{A})$ -closed, cofinal, d -diagonalizing ultrafilters for which for every definable coloring $\pi : [A]^2 \rightarrow 2$ the set*

$$X_\pi := \left\{ a \in A \mid \{b \in A \mid \pi(\{a, b\}) = 1\} \in \mathcal{U} \right\}$$

is definable are d -Ramsey.

Proof. Assume $\mathcal{U}$ is a $\text{cf}(\mathfrak{A})$ -closed, cofinal, d -diagonalizing ultrafilter. Let $\pi : [A]^2 \rightarrow 2$ be defined by $\varphi(x, y)$, let $\sigma(x)$ and $\chi_\pi(x)$ define a cofinal set $S \in \mathcal{U}$ and X_π , respectively, and consider the family $(Y_a)_{a \in A}$ defined by

$$\begin{aligned} \psi(x, y) := & \exists z(\sigma(z) \wedge x \leq z \wedge \forall u((\sigma(u) \wedge u < z) \rightarrow u < x) \\ & \wedge (\chi_\pi(z) \leftrightarrow \varphi(z, y))). \end{aligned}$$

A set Y_a from this family equals $\varphi(\hat{a}, \mathfrak{A})$ for $\hat{a}$ the least element in S greater or equal a if $\varphi(\hat{a}, \mathfrak{A})$ is contained in $\mathcal{U}$, otherwise it equals $A \setminus \varphi(\hat{a}, \mathfrak{A})$, so $(Y_a)_{a \in A}$ is clearly a family in $\mathcal{U}$. Now, define the non-increasing family $(Z_a)_{a \in A}$ by

$$\zeta(x, y) := \sigma(y) \wedge \forall z(z \leq x \rightarrow \psi(z, y)).$$

Notice that this is a family in $\mathcal{U}$ as well because for every $a \in A$ it holds that $Z_a = S \cap \bigcap_{b \leq a} Y_b$ by definition, which is a superset of $S \cap \bigcap_{b \in [a+1] \cap S} Y_b \cap Y_{\hat{a}}$ for $\hat{a}$ the smallest element in S greater or equal to a by the definition of $(Y_a)_{a \in A}$. The latter set is an intersection of less than $\text{cf}(\mathfrak{A})$ many sets in $\mathcal{U}$, hence, it is itself contained in $\mathcal{U}$ by $\text{cf}(\mathfrak{A})$ -closedness.

Now, choose a $Z \in \mathcal{U}$ diagonalizing $(Z_a)_{a \in A}$. Assume without loss of generality that $Z \subseteq S$, otherwise take $Z \cap S$ instead. Let $a \in Z$ and choose $\alpha \in \mathfrak{A}$ such that $s(\alpha) = a$. Then,

$$Z \setminus [a + 1] \subseteq Z_a \subseteq Y_a = Y_{s(\alpha)}.$$

Since Y_a is either $\varphi(a, \mathfrak{A})$ or $A \setminus \varphi(a, \mathfrak{A})$, we get that for all $b, c \in Z \setminus [a + 1]$ it holds that $\pi(\{a, b\}) = \pi(\{a, c\})$. Let $\tau : Z \rightarrow 2$ be the unary coloring of Z mapping each $a \in Z$ to this unique value, i. e.,

$$\tau(a) := \begin{cases} 1 & \text{if there exists a } b \in Z \text{ with } \pi(\{a, b\}) = 1 \\ 0 & \text{otherwise} \end{cases}$$

Since the sets $\{a \in Z \mid \tau(a) = 0\}$ and $\{a \in Z \mid \tau(a) = 1\}$ form a partition of $Z \in \mathcal{U}$ into two pieces, one of them is contained in $\mathcal{U}$. Denote this set by X . It follows that $\tau \upharpoonright X$ has a constant value, denoted by x . Then, $\pi(\{a, b\}) = \tau(a) = x$ for all $a, b \in X$ with $a < b$, so X is homogeneous for π . $\square$

Proposition 2.2.17. *If there exists a definable cofinal set, then for a $\text{cf}(\mathfrak{A})$ -closed, cofinal, d -diagonalizing ultrafilter $\mathcal{U}$ it holds that for any definable partition $\mathcal{P}$ of A either $\mathcal{U} \cap \mathcal{P} \neq \emptyset$ or there exists a set $X \in \mathcal{U}$ such that $|X \cap P| \leq 1$ for all $P \in \mathcal{P}$.*

Proof. This follows from the proof of the above result and the observation that for a given partition $\mathcal{P}$ defined by $\varphi(x, y)$ one can define a corresponding coloring $\pi : [A]^2 \rightarrow 2$ by

$$\psi(x, y) := \exists z(\varphi(x, z) \wedge \varphi(y, z))$$

for which the set X_π is definable. Clearly, $\mathcal{U}$ cannot contain two distinct elements of $\mathcal{P}$. So, either it contains no such element, in which case X_π is empty and thus trivially definable, or it contains a single such element $P \in \mathcal{P}$, yielding $X_\pi = P = \varphi(\mathfrak{A}, a)$ for some $a \in A$. The above proof then yields an unbounded homogeneous set $X \in \mathcal{U}$ for π that either is a subset of some $P \in \mathcal{P}$ if $\pi \upharpoonright X$ is constantly 0, giving $P \in \mathcal{U}$, or satisfies $|X \cap P| \leq 1$ for all $P \in \mathcal{P}$ if $\pi \upharpoonright X$ is constantly 1. $\square$

2.3 d - P -Points and d - Q -Points

This section is concerned with the investigation of two weaker combinatorial ultrafilter properties, namely d - P -points and d - Q -points, and how they relate to d -Ramsey ultrafilters. This is particularly interesting because it is known that an ultrafilter over ω is Ramsey if and only if it is both a P -point and a Q -point.[6, Chapter 11] For the non-standard, definable setting, the forward direction is going to follow from the definitions. The backward direction, however, is likely to be false. Again, fix a countable language L containing the binary relation symbol $<$ and an ordered L -structure $\mathfrak{A}$ modeling $T_{<}$.

Definition 2.3.1. An ultrafilter $\mathcal{U}$ is a d - P -point if for every definable partition $\mathcal{P}$ either $\mathcal{U} \cap \mathcal{P} \neq \emptyset$ or there exists an $X \in \mathcal{U}$ such that $X \cap P$ is bounded for all $P \in \mathcal{P}$.

Definition 2.3.2. An ultrafilter $\mathcal{U}$ is a d - Q -point if for every definable bounded partition $\mathcal{P}$, i. e., a partition into bounded pieces, there exists an $X \in \mathcal{U}$ such that $|X \cap P| \leq 1$ for all $P \in \mathcal{P}$.

Proposition 2.3.3. Every d -selective ultrafilter is a d - P -point and a d - Q -point.

Proof. This follows from the definitions. □

Just like d -selectivity, d - P -points and d - Q -points can be equivalently stated in terms of functions from A to A . This is clear by Definition 2.2.4. The arguments are nonetheless written out below in order to demonstrate the immediate relation between partitions and functions.

Definition 2.3.4. A function $f : A \rightarrow A$ is *bounded-to-one* if $f^{-1}(a) := \{b \in A \mid f(b) = a\}$ is bounded for all $a \in A$.

Proposition 2.3.5. An ultrafilter $\mathcal{U}$ is a d - P -point if and only if for every definable function $f : A \rightarrow A$ there exists a set $X \in \mathcal{U}$ such that $f \upharpoonright X$ is constant or bounded-to-one.

Proof. Let $\varphi(x, y)$ define f . Then, $\varphi(x, y)$ also defines the partition $\mathcal{P}_f$. By the d - P -point property of $\mathcal{U}$, there either exists a set $X \in \mathcal{U} \cap \mathcal{P}_f$ and thus $f \upharpoonright X$ is constant, or there exists a set $X \in \mathcal{U}$ such that $X \cap P$ is bounded for each $P \in \mathcal{P}_f$. Therefore, by the definition of $\mathcal{P}_f$, the set $(f \upharpoonright X)^{-1}(a)$ is bounded for every $a \in A$. Conversely, let $\mathcal{P}$ be some partition defined by $\varphi(x, y)$. Then, $\varphi(x, y)$ defines a function $f_{\mathcal{P}} : A \rightarrow A$ representing $\mathcal{P}$ for which there exists some $X \in \mathcal{U}$ such that $f_{\mathcal{P}} \upharpoonright X$ is constant or bounded-to-one by assumption. X clearly witnesses the d - P -point property for $\mathcal{P}$. □

Proposition 2.3.6. An ultrafilter $\mathcal{U}$ is a d - Q -point if and only if for every definable bounded-to-one function $f : A \rightarrow A$ there exists a set $X \in \mathcal{U}$ such that $f \upharpoonright X$ is injective.

Proof. Let $\varphi(x, y)$ define f . Then, $\varphi(x, y)$ also defines a bounded partition $\mathcal{P}_f$ to which the d - Q -point property can be applied, yielding the set $X \in \mathcal{U}$ with the desired property. Conversely, for a given bounded partition $\mathcal{P}$ defined by $\varphi(x, y)$, the definition corresponds to a bounded-to-one function $f_{\mathcal{P}} : A \rightarrow A$. Choose $X \in \mathcal{U}$ such that $f_{\mathcal{P}} \upharpoonright X$ is injective. Then, there exists a $P \in \mathcal{P}$ with $|X \cap P| \leq 1$ for all $P \in \mathcal{P}$. □

Perhaps the most important property of P -points is their closedness under pseudo-intersections, defined below. In the literature, this is usually the property in terms of which P -points are defined, but it is equivalent to the partition property. The reason for defining d - P -points in terms of partitions, as above, is that the equivalence in the definable settings seems to fail in general and, for the sake of simplicity, the strongest related property is chosen as the definition.

Definition 2.3.7. For sets $X, Y \subseteq A$, X is *almost included* in Y if $X \setminus Y$ is bounded. As a short-hand notation, write $X \subseteq^* Y$. A *pseudo-intersection* of a family $\mathcal{X}$ is a set $Y \subseteq A$ such that $Y \subseteq^* X$ for every $X \in \mathcal{X}$.

Before proceeding, a brief characterization of the relation $\subseteq^*$ is expedient. As proved in Remark 2.3.8, it is both reflexive and transitive, but is clearly not anti-symmetric, since for example the sets $X := A \setminus \{0\}$ and $Y := A \setminus \{1\}$ are not equal, but it holds that $X \subseteq^* Y$ and $Y \subseteq^* X$. Hence, it is a preorder.

Remark 2.3.8. $\subseteq^*$ is a preorder.

Proof. Reflexity is obvious because $X \setminus X = \emptyset$ is bounded for any set $X \subseteq A$. For transitivity assume that $X \subseteq^* Y \subseteq^* Z$ with $X, Y, Z \subseteq A$. Then, it holds that $X \setminus Z = (X \setminus Y \cup X \cap Y) \setminus Z = X \setminus Y \setminus Z \cup (X \cap Y) \setminus Z \subseteq X \setminus Y \cup Y \setminus Z$, which is bounded by assumption, hence $X \subseteq^* Z$. $\square$

The following proposition relates the partition property of d - P -points to the existence of unbounded pseudo-intersections. Note that, again, the existence of a definable cofinal set is assumed. As for the d -Ramsey ultrafilters, this assumption can be omitted when focussing on robust d - P -points, i. e., ultrafilters that remain d - P -points in some expansion $(\mathfrak{A}, S)$ for a cofinal set S being the interpretation of a new unary relation symbol. This again works because the d - P -point property is clearly preserved under taking reducts.

Proposition 2.3.9. Let $\mathcal{U}$ be a d - P -point. If there exists a definable cofinal set, then $\mathcal{U}$ contains a pseudo-intersection for every definable non-increasing chain $(X_a)_{a \in A}$ in $\mathcal{U}$.

Proof. Let $s : \text{cf}(\mathfrak{A}) \rightarrow A$ be unbounded and increasing with $\text{im}(s)$ definable by $\sigma(x)$, let $\chi(x, y)$ define $(X_a)_{a \in A}$. It suffices to find a pseudo-intersection for $(X_a)_{a \in \text{im}(s)}$. Consider $Z := \bigcap_{a \in \text{im}(s)} X_a$. If $Z \in \mathcal{U}$, we are done, so assume $Z \notin \mathcal{U}$. Let

$$\begin{aligned} \pi(x, y) := & \sigma(y) \wedge (y = s(0) \wedge (\forall z \chi(z, x) \vee \forall z \neg \chi(z, x)) \\ & \vee y \neq s(0) \wedge \neg \chi(y, x) \wedge \forall z < y (\sigma(z) \rightarrow \chi(z, x))) \end{aligned}$$

define the partition $\mathcal{P} := \{Y_a \mid a \in \text{im}(s)\}$ with $Y_a := \pi(\mathfrak{A}, a)$. Note that $\pi(\mathfrak{A}, s(0)) = Z \cup (A \setminus X_0)$ and $\pi(\mathfrak{A}, a) = \bigcap_{b \in \text{im}(s) \cap [a]} X_b \setminus X_a$ for $a \in \text{im}(s) \setminus \{s(0)\}$. Let $X \in \mathcal{U}$ have a bounded intersection with every Y_a . Such an X exists because $\mathcal{U}$ is a d - P -point and $Y_a \notin \mathcal{U}$ for all $a \in \text{im}(s)$. Then, X is a pseudo-intersection of $(X_a)_{a \in \text{im}(s)}$. For a given $a \in \text{im}(s)$, it is to show that $X \subseteq^* X_a$. Write $X = \bigcup_{b \in \text{im}(s)} (Y_b \cap X)$. For every $b > a$

it holds that $Y_b \cap X \subseteq Y_b \subseteq X_a$, so it suffices to show that $\bigcup_{b \in [a+1] \cap \text{im}(s)} (Y_b \cap X)$ is bounded. For each $b \in [a+1] \cap \text{im}(s)$, let c_b be a bound of $Y_b \cap X$. Then, the set of these bounds is bounded because $|[a+1] \cap \text{im}(s)| < \text{cf}(\mathfrak{A})$. Denote its bound by d . It follows that $\bigcup_{b \in [a+1] \cap \text{im}(s)} (Y_b \cap X) \subseteq [d]$. $\square$

Since we know from the standard theory that being a P -point is equivalent to being closed under pseudo-intersections, we ask what can be proved from the pseudo-intersection property in the non-standard setting. It follows a list of results proving, under various assumptions, from the closedness under pseudo-intersections different partition properties related to d - P -points, starting with Proposition 2.3.10, which shows the d - P -point property for a quite restricted set of partitions. However, its proof can be used to show a similar result for a more general set when assuming the ordered structure $\mathfrak{A}$ to be κ -like. This notion is defined below. When assuming the investigated ultrafilter to be $\text{cf}(\mathfrak{A})$ -closed, this slightly more general partition property can in turn be further generalized to the d - P -point property restricted to all partitions with at most $\text{cf}(\mathfrak{A})$ many pieces.

Proposition 2.3.10. *Let $f : A \rightarrow A$ be an unbounded, definable function with image $\text{im}(f)$ of order type $\text{cf}(\mathfrak{A})$, and let $\mathcal{P}_f$ be the partition it represents. If an ultrafilter $\mathcal{U}$ contains a pseudo-intersection for every definable non-increasing chain $(X_a)_{a \in A}$ in $\mathcal{U}$, then there either exists a $\mathcal{Q} \subseteq \mathcal{P}_f$ with $|\mathcal{Q}| < \text{cf}(\mathfrak{A})$ such that $\bigcup \mathcal{Q} \in \mathcal{U}$ or there exists an $X \in \mathcal{U}$ such that $X \cap P$ is bounded for every $P \in \mathcal{P}_f$. Moreover, the first case holds for every partition with less than $\text{cf}(\mathfrak{A})$ many pieces.*

Proof. Clearly, if a partition $\mathcal{P}$ has size less than $\text{cf}(\mathfrak{A})$, then $\bigcup \mathcal{P} = A \in \mathcal{U}$, so the first case trivially holds. Now consider the partition $\mathcal{P}_f$ and assume the first case does not occur. In particular, $\mathcal{P}_f \cap \mathcal{U} = \emptyset$. We have to verify the second case. Let $\varphi(x, y)$ define the function f and write $\mathcal{P}_f$ as $\{P_b \mid b \in \text{im}(f)\}$ with $P_b = f^{-1}(b)$. For every $a \in A$ define

$$X_a := \bigcup \{P_b \mid a <^{\mathfrak{A}} b\}.$$

Then $(X_a)_{a \in A}$ is a non-increasing chain definable by

$$\chi(x, y) := \exists z (x < z \wedge \varphi(y, z)).$$

Then, $(X_a)_{a \in A}$ is in $\mathcal{U}$. Otherwise, there is an $a \in A$ such that $X_a \notin \mathcal{U}$, so $A \setminus X_a = \bigcup \{P_b \mid b \leq a\} \in \mathcal{U}$ is the union of less than $\text{cf}(\mathfrak{A})$ many sets in $\mathcal{P}_f$, verifying the first case.

Let $X \in \mathcal{U}$ be a pseudo-intersection of $(X_a)_{a \in A}$. This witnesses the second case. Otherwise, there is an $a \in \text{im}(f)$ such that $X \cap P_a$ is unbounded. Then, $P_a \subseteq A \setminus X_a$, so $X \setminus X_a \supseteq X \cap P_a$ is unbounded, contradicting the choice of X . $\square$

As mentioned above, in order to get a more general partition property, the notion of κ -likeness is useful. The goal is to omit the condition that the partitions have to be represented by some definable unbounded function with well-ordered image. The purpose of this requirement in the above proof is to ensure that either the family $(X_a)_{a \in A}$

is indeed a family in $\mathcal{U}$, or there exists a set in $\mathcal{U}$ that is the union of less than $\text{cf}(\mathfrak{A})$ many pieces of $\mathcal{P}$. However, as will be shown in the proof of Proposition 2.3.14, this is also a consequence of the condition that every proper initial segment of $\mathfrak{A}$ is smaller than $\text{cf}(\mathfrak{A})$, which in turn is closely related to κ -likeness.

Definition 2.3.11. A model of size κ is κ -like if every proper initial segment has cardinality less than κ .

Remark 2.3.12. $\mathfrak{A}$ is κ -like and $\text{cf}(\mathfrak{A}) = \kappa$ if and only if every proper initial segment of $\mathfrak{A}$ has cardinality less than $\text{cf}(\mathfrak{A})$.

Proof. The direction from left to right is clear by assumption and the definition of κ -likeness. On the other hand, the right-hand side implies κ -likeness since $\text{cf}(\mathfrak{A}) \leq \kappa$. Furthermore, A is the union of $\text{cf}(\mathfrak{A})$ many pieces of size less than $\text{cf}(\mathfrak{A})$, thus, if $\text{cf}(\mathfrak{A}) < \kappa$, then κ can be bounded from above by $\text{cf}(\mathfrak{A})^2 = \text{cf}(\mathfrak{A}) < \kappa$, leading to a contradiction. $\square$

By the following remark, the assumption that $\text{cf}(\mathfrak{A}) = \kappa$ holds for all κ -like ordered structures $\mathfrak{A}$ in the case that κ is regular. Nonetheless, it might also hold for certain κ -like structures with singular κ .

Remark 2.3.13. If $\mathfrak{A}$ is κ -like, then $\text{cf}(\mathfrak{A}) \geq \text{cf}(\kappa)$. In particular, $\text{cf}(\mathfrak{A}) = \kappa$ for regular κ .

Proof. Assume for a contradiction that $\text{cf}(\mathfrak{A}) < \text{cf}(\kappa)$. Let $s : \text{cf}(\mathfrak{A}) \rightarrow A$ be unbounded and increasing. If $\lambda := \bigcup_{\alpha \in \text{cf}(\mathfrak{A})} |[s(\alpha)]|$ is smaller than κ , the size of A can be bounded from above by $\lambda \cdot \text{cf}(\mathfrak{A}) = \max\{\lambda, \text{cf}(\mathfrak{A})\} < \kappa$, contradicting that A is of size κ . Consequently, $\lambda = \kappa$, and thus $\{|[s(\alpha)]| \mid \alpha \in \text{cf}(\mathfrak{A})\}$ is an unbounded subset of κ , contradicting the assumption that $\text{cf}(\mathfrak{A}) < \text{cf}(\kappa)$. The additional remark is then clear by $\kappa = \text{cf}(\kappa) \leq \text{cf}(\mathfrak{A}) \leq \kappa$. $\square$

Having some understanding of how the cofinality of κ -like structures behaves, it is now clear that assuming $\mathfrak{A}$ to be κ -like such that $\text{cf}(\mathfrak{A}) = \kappa$ is sufficient to ensure that every initial segment of $\mathfrak{A}$ has size less than $\text{cf}(\mathfrak{A})$, simplifying the required assumptions about the partitions, as noted above. See the following proposition.

Proposition 2.3.14. If $\mathfrak{A}$ is κ -like and $\text{cf}(\mathfrak{A}) = \kappa$, then for every ultrafilter $\mathcal{U}$ that contains a pseudo-intersection for every definable non-increasing chain $(X_a)_{a \in A}$ in $\mathcal{U}$, it holds that for every definable partition $\mathcal{P}$ with cardinality less or equal to $\text{cf}(\mathfrak{A})$ there either is $\mathcal{Q} \subseteq \mathcal{P}$ with $|\mathcal{Q}| < \text{cf}(\mathfrak{A})$ such that $\bigcup \mathcal{Q} \in \mathcal{U}$ or there is $X \in \mathcal{U}$ such that $X \cap P$ is bounded for every $P \in \mathcal{P}$.

Proof. The argument in the proof of Proposition 2.3.10 still holds using the modified assumptions. Observe that, if $\mathcal{P}$ has size $\text{cf}(\mathfrak{A})$, the function $f_{\mathcal{P}}$ representing $\mathcal{P}$ is unbounded by the choice of $\mathfrak{A}$ and Remark 2.3.12. Then, $(X_a)_{a \in A}$ is in $\mathcal{U}$ as otherwise for any $X_a \notin \mathcal{U}$ the set $A \setminus X_a = \bigcup_{b \leq a} P_b \in \mathcal{U}$ is the union of less than $\text{cf}(\mathfrak{A})$ many sets in $\mathcal{P}$ since it is indexed by a bounded set, which, again by the choice of $\mathfrak{A}$, has size less than $\text{cf}(\mathfrak{A})$. $\square$

Of course this weaker partition property can be rephrased in terms of functions in the intuitive way.

Proposition 2.3.15. *If an ultrafilter $\mathcal{U}$ has the above partition property, then for every definable function $f : A \rightarrow A$ with $|im(f)| \leq cf(\mathfrak{A})$ there exists a set $X \in \mathcal{U}$ such that $f|X$ is either bounded-to-one or bounded.*

Proof. Let $\varphi(x, y)$ define f . Then, $\varphi(x, y)$ also defines the partition $\mathcal{P}_f$ to which the partition property can be applied, yielding the set X with the desired properties. $\square$

Finally, assuming that $\mathcal{U}$ is $cf(\mathfrak{A})$ -closed, one can complete the circle of implications by showing that the weak partition property implies the d - P -point property restricted to partitions $\mathcal{P}$ with $|\mathcal{P}| \leq cf(\mathfrak{A})$. It should however be remarked that the existence of a $cf(\mathfrak{A})$ -closed ultrafilter for uncountable $cf(\mathfrak{A})$ implies the existence of measurable cardinals and is therefore not provable in ZFC. See the comments in Section 3.4.

Proposition 2.3.16. *For every $cf(\mathfrak{A})$ -closed ultrafilter and every definable partition $\mathcal{P}$ with cardinality less or equal to $cf(\mathfrak{A})$ containing a subset $\mathcal{Q}$ with $|\mathcal{Q}| < cf(\mathfrak{A})$ such that $\bigcup \mathcal{Q} \in \mathcal{U}$, there exists a $Q \in \mathcal{P} \cap \mathcal{U}$.*

Proof. This is clear by $cf(\mathfrak{A})$ -closedness of $\mathcal{U}$. $\square$

2.4 Strong Variants

As we have seen in the previous section, if an ultrafilter is d -selective, it is both a d - P -point and a d - Q -point. Contrary to the standard setting in which being both a P -point and a Q -point implies being selective, and thus Ramsey, the converse is not clear. However, it becomes true when strengthening the d - P -point property by requiring it not to contain arbitrary homogeneous sets for definable partitions but definable such sets. Since undefinable sets become obsolete for such considerations, this leads to the investigations of combinatorial ultrafilters over the Boolean algebra of definable sets.

Definition 2.4.1. Let $L_{BA} := \{\cup, \sim, 1\}$ be the language of Boolean algebras. The *Boolean algebra $\mathfrak{D}(\mathfrak{A})$ of definable sets* is the L_{BA} -structure $(D(\mathfrak{A}), \cup^{\mathfrak{D}(\mathfrak{A})}, \sim^{\mathfrak{D}(\mathfrak{A})}, A)$ where $D(\mathfrak{A})$ denotes the set of definable sets in the structure $\mathfrak{A}$ and $\cup^{\mathfrak{D}(\mathfrak{A})}$ and $\sim^{\mathfrak{D}(\mathfrak{A})}$ are the usual union and complement operations.

This structure is indeed a Boolean algebra since for arbitrary $L(A)$ -formulas φ, ψ it holds that

$$\begin{aligned} A &= (x = x)(\mathfrak{A}), \\ \varphi(\mathfrak{A}) \cup \psi(\mathfrak{A}) &= (\varphi \vee \psi)(\mathfrak{A}), \\ \sim \varphi(\mathfrak{A}) &= (\neg \varphi)(\mathfrak{A}), \end{aligned}$$

so the satisfaction of the necessary axioms transfers from the power set algebra. Now, in order to terminologically distinguish the usual ultrafilters over $\mathfrak{A}$ from ultrafilters over $\mathfrak{D}(\mathfrak{A})$, the following term is introduced.

Definition 2.4.2. A d -ultrafilter is an ultrafilter over $\mathfrak{D}(\mathfrak{A})$.

All the combinatorial ultrafilter properties can now be defined identically for d -ultrafilters, i. e., a d -ultrafilter is d -Ramsey if it contains a homogeneous set for every definable binary coloring $\pi : [A]^2 \rightarrow 2$, etc. Of course such a homogeneous set is definable since the universe $D(\mathfrak{A})$ does not even contain undefinable sets. Whether or not $D(\mathfrak{A})$ contains homogeneous sets is entirely dependent on the structure $\mathfrak{A}$. The strengthening of the combinatorial ultrafilter properties towards the inclusion of definable special sets, as alluded to above, can now be defined in the obvious sense by simply extending d -ultrafilters with the related properties.

Definition 2.4.3. An ultrafilter is *strongly d -Ramsey* if it extends a d -Ramsey d -ultrafilter. It is *strongly d -selective* if it extends a d -selective d -ultrafilter. It is a *strong d - P -point* if it extends a d -ultrafilter with the d - P -point property. It is a *strong d - Q -point* if it extends a d -ultrafilter with the d - Q -point property.

The following proposition then shows that being both a strong d - P -point and a d - Q -point is sufficient for an ultrafilter to be d -selective by first applying to a given definable partition the strong d - P -point property, yielding some definable set that can be used to define a bounded partition, for which the ultrafilter contains a homogeneous set by the d - Q -point property. The main advantage over the regular d - P -point property is that this intermediate bounded partition becomes definable.

Proposition 2.4.4. *If an ultrafilter $\mathcal{U}$ is both a strong d - P -point and a d - Q -point, then it is d -selective.*

Proof. Let $\mathcal{P}$ be a partition defined by $\varphi(x, y)$ and assume that $\mathcal{U} \cap \mathcal{P} = \emptyset$. Since $\mathcal{U}$ is a d - P -point, there is an $X \in \mathcal{U}$ such that $X \cap P$ is bounded for every $P \in \mathcal{P}$, say it is defined by $\chi(x)$. Let $\mathcal{Q}$ be the partition defined by

$$\psi(x, y) := \neg\chi(x) \wedge x=y \vee \chi(x) \wedge \varphi(x, y).$$

Then, $\mathcal{Q}$ is bounded. Since $\mathcal{U}$ is a d - Q -point, there is a $Y \in \mathcal{U}$ such that $|Y \cap Q| \leq 1$ for every $Q \in \mathcal{Q}$. Hence, $Z := X \cap Y$ is in $\mathcal{U}$ and $|Z \cap P| \leq 1$ for every $P \in \mathcal{P}$. $\square$

When additionally requiring the ultrafilter to be a strong d - Q -point, this yields strong d -selectivity. Hence, an ultrafilter is strongly d -selective if and only if it is both a strong d - P -point and a strong d - Q -point.

Corollary 2.4.5. *If an ultrafilter $\mathcal{U}$ is both a strong d - P -point and a strong d - Q -point, then it is strongly d -selective.*

Proof. The proof is identical to the previous one except for choosing Y in the last step as a definable set. $\square$

When addressing the question if d -selectivity or its strong variant imply that an ultrafilter is d -Ramsey or even $d^{<\omega}$ -Ramsey, it becomes difficult to find a definite answer. Yet, there is evidence that this is generally not the case. The main problem seems to

be the fact that such proofs require, informally speaking, a certain definable access to the ultrafilter. An example is provided by the proof of Proposition 2.2.16, which closely follows an analogous proof in [6, Chapter 11] showing that any selective ultrafilter over ω is Ramsey using diagonalization. However, in order to apply d -diagonalization, one has to find a special uniformly definable family of sets inside the ultrafilter. This can be done when assuming the definability of the sets X_π , which provides exactly the required kind of definable access to the ultrafilter, but there is no reason to suppose that something similar is possible without such additional assumptions. More generally, this problem constitutes the primary obstacle for showing that a d^n -Ramsey ultrafilter is d^{n+1} -Ramsey and makes an equivalence of d -Ramsey and $d^{<\omega}$ -Ramsey unlikely.

3 Existence

This chapter comprises an extensive list of results addressing the question of existence of combinatorial ultrafilters, although the main focus will be on d -Ramsey ultrafilters because they imply the existence of d - P -points and d - Q -points. Further, this restriction to d^2 -Ramsey ultrafilters is for the sake of simplicity and without loss of generality. All of the following existence proofs are completely analogous for d^n -Ramsey ultrafilters with $n > 2$ and even $d^{<\omega}$ -Ramsey ultrafilters because the only required property of definable binary colorings is that there are only few of them, which clearly holds as well for the set of arbitrary definable colorings. The results are grouped by the class of ordered structures to which they apply, starting with the most restrictive case of countable structures, continuing with a more generalized notion of small structures, and concluding with a section on uncountable structures. Finally, an attempt is made to separate the different ultrafilter properties. For the rest of the chapter, fix a countable language L containing the binary relation symbol $<$ and an ordered L -structure $\mathfrak{A}$ satisfying the theory $T_{<}$ from Definition 2.1.7.

3.1 Countable Ordered Structures

One major advantage of countable structures for proving the existence of d -Ramsey ultrafilters is that every coloring of the universe A has a homogeneous set which is unbounded with regards to $<^{\mathfrak{A}}$. In other words, if $\mathfrak{A}$ is countable, it has the following property.

Definition 3.1.1. The order $<^{\mathfrak{A}}$ *admits homogeneous sets* if for every coloring $\pi : [A]^2 \rightarrow 2$ there exists an unbounded homogeneous set $H \subseteq A$.

As the following lemma shows, every binary coloring of a countable $\mathfrak{A}$ has an unbounded homogeneous set, which can be proved simply by applying Ramsey's theorem over a cofinal set. Admitting homogeneous sets is in fact a crucial property for proving the existence of d -Ramsey ultrafilters by an abstract argument. If the order $<^{\mathfrak{A}}$ did not have this property, i. e., if there was some binary coloring having no unbounded homogeneous set, then expanding $\mathfrak{A}$ by a relation symbol defining this coloring yields an ordered structure over which no d -Ramsey ultrafilter exists. But this structure differs only in the language it interprets, hence, the abstract approach making no assumptions about the language L , beside its countability, and how it is interpreted has reached a limit that can only be passed by making more specialized assumptions. Such assumptions, however, equate concrete arithmetic theories and are, with a few exceptions, not investigated in this thesis.

Lemma 3.1.2. *If $\mathfrak{A}$ is countable, then every binary coloring has an unbounded homogeneous set $X \subseteq A$.*

Proof. Let a coloring $\pi : [A]^2 \rightarrow 2$ be given. Since $\mathfrak{A}$ is countable, $\text{cf}(\mathfrak{A})$ is $\aleph_0$. Take a cofinal set X . Then, by Theorem 2.1.2, there exists a homogeneous set $Y \subseteq X$ with $|Y| = \aleph_0$ for the coloring $\pi \upharpoonright [X]^2$. This set is clearly an unbounded set with respect to the order $<^{\mathfrak{A}} \cap X^2$ and, by unboundedness of X , is unbounded in A . $\square$

Having verified the existence of unbounded homogeneous sets for countable $\mathfrak{A}$, in order to prove the existence of d -Ramsey ultrafilters, one has to show that there is a family with the unbounded finite intersection property containing such a set for each definable coloring. This is where a striking difference between the Ramsey and d -Ramsey properties becomes apparent. Such a family is required to include one special set for each definable binary coloring, of which only $|L| \cdot |A|$ many exist. Thus, assuming the countability L , there are only $|A|$ many sets to include, whereas proving the existence of a Ramsey ultrafilter demands the inclusion of $2^{|A|}$ many sets and as a consequence depends on additional set-theoretic axioms. The proof of the following proposition claiming the existence of d -Ramsey ultrafilters over countable ordered structures is analogous to the proof of Theorem 2.1.6 with the crucial difference that the chain construction does not require any limit steps, making it arguably the simplest possible case.

Proposition 3.1.3. *Let $\mathfrak{A}$ be countable. Then, there exists a d -Ramsey ultrafilter over $\mathfrak{A}$.*

Proof. Let $\{\pi_n \mid n \in \aleph_0\}$ be the set of all definable binary colorings over $\mathfrak{A}$. Let X_0 be an unbounded homogeneous set for π_0 and assume X_n with $n \in \aleph_0$ is defined to be an unbounded set. Choose for X_{n+1} an unbounded homogeneous set of $\pi_{n+1} \upharpoonright [X_n]^2$. Then, the family $\{X_n \mid n \in \aleph_0\}$ has the unbounded finite intersection property and is therefore contained in an unbounded ultrafilter $\mathcal{U}$, which is d -Ramsey by construction. $\square$

This result can for example be strengthened by additionally including a cofinal set in the chain construction, which proves the existence of cofinal d -Ramsey ultrafilters over countable ordered structures. This proof is included below for demonstration, but this is also an immediate consequence of Corollary 2.2.11 stating that robustly d -Ramsey ultrafilters are cofinal and the fact that, as alluded to in Section 2.2, robustly d -Ramsey ultrafilters exist for the class of countable ordered structures. This follows from Proposition 3.1.3 and the observation that every expansion $(\mathfrak{A}, S)$ for a cofinal set S is clearly a countable ordered structure interpreting a countable language, making Proposition 3.1.3 applicable.

Corollary 3.1.4. *If $\mathfrak{A}$ is countable, then there exists a cofinal d -Ramsey ultrafilter.*

Proof. This can be proved analogously to Proposition 3.1.3, except by defining X_0 as an unbounded homogeneous set for the coloring $\pi_0 \upharpoonright [\text{im}(s)]^2$ for some unbounded, increasing function $s : \text{cf}(\mathfrak{A}) \rightarrow A$. The resulting ultrafilter is again unbounded and d -Ramsey but additionally contains the cofinal set $\text{im}(s)$. $\square$

The following results on the existence of strongly d -Ramsey ultrafilters and strong d - P -points conclude the investigation of countable structures. It is important to note that the structure constructed in the proof of Proposition 3.1.5 below can hardly be considered of practical value. Rather, this proposition serves the sole purpose of proving the consistency of the notion of strongly d -Ramsey ultrafilters. More extensive results in this direction require a certain knowledge about which sets can be defined in a given structure, which, as mentioned above, is outside the scope of this thesis.

Proposition 3.1.5. *Let $\mathfrak{A}$ be countable. Then, it has an L' -expansion $\mathfrak{A}'$ for a countable $L' \supseteq L$ for which a strongly d -Ramsey ultrafilter exists.*

Proof. Define the expansion $\mathfrak{A}'$ inductively and in parallel construct an almost non-increasing chain of unbounded homogeneous sets that is going to be extended to a strongly d -Ramsey ultrafilter. Let $\mathfrak{A}_0 := \mathfrak{A}$, $L_0 := L$, Z_0 be some cofinal set, assume for $n \in \omega$ that $\mathfrak{A}_n$ is defined and interprets the language L_n , and that Z_n is again a cofinal set. Let $(\pi_{n,0}, X_{n,0}), (\pi_{n,1}, X_{n,1}), \dots$ enumerate pairs of all binary colorings definable in $\mathfrak{A}_n$, each paired with an unbounded set such that $Z_n \supseteq X_{n,0} \supseteq X_{n,1} \supseteq \dots$ and $X_{n,i}$ is homogeneous for $\pi_{n,i}$. This chain of sets can be constructed in exactly the same way as in the proof of Proposition 3.1.3. Further, let $R_n := \{R_{n,m} \mid m \in \omega\}$ be a countable set of new unary relation symbols with $L_n \cap R_n = \emptyset$. Now, define $L_{n+1} := L_n \cup R_n$, expand $\mathfrak{A}_n$ to an L_{n+1} -structure with $R_{n,m}^{\mathfrak{A}_{n+1}} = X_{n,m}$ for all $m \in \omega$ and choose for $Z_{n+1} \subseteq Z_n$ an unbounded pseudo-intersection of $\{X_{n,m} \mid m \in \omega\}$, which exists because this family has the unbounded finite intersection property. Finally, set $L' := \bigcup_{n \in \omega} L_n$ and $\mathfrak{A}' := \bigcup_{n \in \omega} \mathfrak{A}_n$ and observe that a coloring $\pi : [A]^2 \rightarrow 2$ that is definable in $\mathfrak{A}'$ is also definable in $\mathfrak{A}_n$ for some $n \in \omega$, hence, there exists a constant symbol $S \in L_{n+1} \subseteq L'$ such that $S^{\mathfrak{A}'} = S^{\mathfrak{A}_{n+1}}$ is an unbounded homogeneous set for π . Consequently, every coloring definable in $\mathfrak{A}'$ has an unbounded homogeneous set in the family of definable sets $\{X_{n,m} \mid n, m \in \omega\}$. Furthermore, this family has the unbounded finite intersection property by construction, thus, it can be extended to a d -Ramsey d -ultrafilter, which in turn can be extended to a strongly d -Ramsey ultrafilter. $\square$

On the other hand, it can easily be shown that every countable model has a strong d - P -point. The reason for this is that, given an unbounded, definable set and a definable partition, it is straightforward to find a definable subset witnessing the d - P -point property for the partition. This is not the case for the chain construction for d -Ramsey ultrafilters as the set chosen during a successor step comes from Ramsey's theorem and is therefore not necessarily definable.

Proposition 3.1.6. *Let $\mathfrak{A}$ be countable. Then, there exists a strong d - P -point over $\mathfrak{A}$.*

Proof. Let $\{\mathcal{P}_n \mid n \in \mathbb{N}_0\}$ enumerate all definable partitions of A . If there exists an unbounded piece $P \in \mathcal{P}$, then define $X_0 := P$, otherwise choose $X_0 := A$, which obviously has bounded intersection with each piece of $\mathcal{P}_0$. In both cases, X_0 is definable. Analogously, assume X_n with $n \in \mathbb{N}_0$ is an unbounded, definable set. Again, if there exists a piece $P \in \mathcal{P}_{n+1}$ such that $X_n \cap P$ is unbounded, choose $X_{n+1} := X_n \cap P$, which is definable by the conjunction of the formulas defining X_n and P , otherwise $X_{n+1} := X_n$,

witnessing the d - P -point property for $\mathcal{P}_{n+1}$. Then, the family $\{X_n \mid n \in \aleph_0\}$ has the unbounded finite intersection property and is therefore contained in an unbounded ultrafilter $\mathcal{U}$, which is a strong d - P -point by construction. $\square$

3.2 Small Ordered Structures

While the countable case is easy because it allows the construction of non-increasing chains of homogeneous sets with the unbounded finite intersection property, this approach is generally infeasible for larger structures due to the fact that such a construction would have to choose for every limit step a subset of the intersection of all previous sets. This intersection, however, cannot be shown to be non-empty, since the only information known about the previous sets is that they are unbounded. Fortunately, almost non-increasing chains are sufficient.

Remark 3.2.1. *Let I be a linear order, $(X_i)_{i \in I}$ an almost non-increasing chain of unbounded sets, i. e., $X_i \supseteq^* X_j$ for $i < j \in I$. Then, $(X_i)_{i \in I}$ has the unbounded finite intersection property.*

Proof. Proceed by induction over the number of sets and note that the intersection of a single unbounded set is trivially unbounded. Now, assume that the intersection of n elements from the chain with $n \in \omega$ is unbounded and let $Y_0 \supseteq^* \dots \supseteq^* Y_n$ be $n+1$ elements. By the induction hypothesis, $Y := \bigcap_{i \in \{1, \dots, n\}} Y_i$ is unbounded, and it holds that $Y \subseteq Y_1 \subseteq^* Y_0$ and thus $Y \subseteq^* Y_0$. Consequently, since Y is unbounded but $Y \setminus Y_0$ is bounded, $Y \cap Y_0 = \bigcap_{i \in \{0, \dots, n\}} Y_i$ is unbounded as well, concluding the induction step. $\square$

This remark alleviates the constraints on the chain construction in the sense that sets for limit steps are not necessarily required to be subsets of the intersection of the previous sets anymore but subsets of an unbounded pseudo-intersection. Thus it remains to understand under which circumstances such pseudo-intersections exist. We need the following definition.

Definition 3.2.2. A family $\mathcal{X} \subseteq [\kappa]^\kappa$ has the *strong finite intersection property* if for every finite subfamily $\mathcal{Y} \subseteq \mathcal{X}$ it holds that $|\bigcap \mathcal{Y}| = \kappa$. The κ -*pseudo-intersection number* $\mathfrak{p}_\kappa$ is the smallest cardinal such that there exists a family $\mathcal{X} \subseteq [\kappa]^\kappa$ with the strong finite intersection property that has no pseudo-intersection of size κ . Denote $\mathfrak{p}_{\aleph_0}$ by $\mathfrak{p}$.

Clearly, size alone is not helpful for unbounded families over $\mathfrak{A}$ since $<^{\mathfrak{A}}$ might permit bounded sets of size $|A|$, but one can of course start the chain construction with a cofinal set, the order type of which is a regular cardinal, where large sets are exactly the unbounded ones. Hence, if there are at most $\mathfrak{p}_{\text{cf}(\mathfrak{A})}$ definable colorings, every proper initial segment of the chain to be constructed has an unbounded pseudo-intersection, facilitating its completion. This suggests the following class of ordered structures.

Definition 3.2.3. The ordered structure $\mathfrak{A}$ is *small* if $\text{cf}(\mathfrak{A}) = \aleph_0$ and $|A| \leq \mathfrak{p}$.

This is a strict generalization of countable ordered structures because it is known that $\aleph_1 \leq \mathfrak{p} \leq 2^{\aleph_0}$. The following proposition shows that such structures admit d -Ramsey ultrafilters and demonstrates that pseudo-intersections are indeed a useful tool for overcoming the limit step in the chain construction.

Theorem 3.2.4. *If $\mathfrak{A}$ is small, then there exists a d -Ramsey ultrafilter.*

Proof. Let $\kappa := |A|$, let S be a cofinal set, and let $\{\pi_\alpha \mid \alpha \in \kappa\}$ be the set of all definable binary colorings of $\mathfrak{A}$. Choose for $X_0 \subseteq S$ an unbounded homogeneous set for $\pi_0 \upharpoonright [S]^2$ and assume X_α with $\alpha \in \kappa$ is defined to be an unbounded set. Then, choose for $X_{\alpha+1}$ an unbounded homogeneous set of $\pi_{\alpha+1} \upharpoonright [X_\alpha]^2$. If $\lambda \in \kappa$ is a limit ordinal, observe that $\mathcal{X}_\lambda := \{X_\alpha \mid \alpha < \lambda\}$ is a family of infinite subsets of S , which is an isomorphic copy of $\aleph_0$, and it holds that $|\mathcal{X}_\lambda| < \kappa \leq \mathfrak{p}$, where the last inequality follows by the smallness of $\mathfrak{A}$. Hence, since $\mathcal{X}_\lambda$ has the unbounded finite intersection property, it has an unbounded pseudo-intersection Y_λ , and $X_\lambda \subseteq Y_\lambda$ can be chosen as before. Then, the family $\{X_\alpha \mid \alpha \in \kappa\}$ has the unbounded finite intersection property and is therefore contained in an unbounded ultrafilter $\mathcal{U}$, which is d -Ramsey by construction. $\square$

It now seems natural to generalize small structures even further by just requiring $\mathfrak{A}$ to satisfy that $|A| \leq \mathfrak{p}_{\text{cf}(\mathfrak{A})}$ while allowing arbitrary cofinalities. Unfortunately, such a generalization is going to prove unsuccessful due to the following proposition and its subsequent corollary.

Proposition 3.2.5. *For each cardinal κ , if $\text{cf}(\kappa) > \aleph_0$, then $\mathfrak{p}_\kappa = \aleph_0$.*

Proof. For $n \in \omega$, define $Y_n := \{\alpha \in \kappa \mid \text{there exists a limit ordinal } \lambda \text{ with } \alpha = \lambda + n\}$ and $X_n := \bigcup_{\alpha \geq n} Y_\alpha$. Then, $\mathcal{X} := \{X_n \mid n \in \omega\} \subseteq [\kappa]^\kappa$ is countable and has the strong finite intersection property. It remains to show that $\mathcal{X}$ does not have a pseudo-intersection of size κ . Assume for a contradiction that such a pseudo-intersection X exists. Then, $X \setminus X_n$ is bounded for each $n \in \omega$. Denote this bound by β_n . Since the set of these bounds is countable and $\omega < \text{cf}(\kappa)$, it is itself bounded by some $\beta \in \kappa$. But this implies that $X \setminus \beta = \emptyset$, thus X is bounded, contradicting the choice of X . $\square$

Corollary 3.2.6. *There exists no ordered structure $\mathfrak{A}$ with $\text{cf}(\mathfrak{A}) > \aleph_0$ and $|A| \leq \mathfrak{p}_{\text{cf}(\mathfrak{A})}$.*

Proof. Assume such a structure $\mathfrak{A}$ exists. Since $\text{cf}(\mathfrak{A})$ is regular and uncountable, it follows by Proposition 3.2.5 that $\mathfrak{p}_{\text{cf}(\mathfrak{A})} = \aleph_0$ and thus $\text{cf}(\mathfrak{A}) \leq |A| = \aleph_0$, contradicting the assumption that $\text{cf}(\mathfrak{A})$ is uncountable. $\square$

Depending on the mathematical context in which one plans to apply a d -Ramsey ultrafilter, particularly during an ultrapower construction, it might be desirable that it contains additional sets aside from the homogeneous sets. It is not necessarily sufficient for this family to have the unbounded finite intersection property in order to be extendible to a d -Ramsey ultrafilter. However, having an unbounded pseudo-intersection allows to simply prepend to the chain construction in Theorem 3.2.4 some preceding steps enforcing the inclusion of the additional sets. See the following corollary.

Corollary 3.2.7. *Let $\mathfrak{A}$ be small. If a family $\mathcal{X}$ of unbounded subsets of A has an unbounded pseudo-intersection, it can be extended to a d -Ramsey ultrafilter.*

Proof. This can be proved by choosing the unbounded set of order type $\text{cf}(\mathfrak{A})$ in the proof of Theorem 3.2.4 such that it is a subset of the unbounded pseudo-intersection of $\mathcal{X}$ and again construct the family $\{X_\alpha \mid \alpha \in \kappa\}$. Then, $\mathcal{X} \cup \{X_\alpha \mid \alpha \in \kappa\}$ clearly has the unbounded finite intersection property and can therefore be extended to an unbounded d -Ramsey ultrafilter containing $\mathcal{X}$. $\square$

In particular, if a given countable family has the unbounded finite intersection property, it always has an unbounded pseudo-intersection, provided that the structure has countable cofinality as is the case in small ordered structures.

Corollary 3.2.8. *Let $\mathfrak{A}$ be small and let $(X_n)_{n \in \omega}$ be a countable family of unbounded subsets of A with the unbounded finite intersection property. Then, $\mathcal{X}$ can be extended to a d -Ramsey ultrafilter.*

Proof. By the previous corollary, it suffices to show that $\mathcal{X}$ has an unbounded pseudo-intersection. Let $s : \aleph_0 \rightarrow A$ be an unbounded, increasing function and let a_0 be any element of X_0 . Now, inductively define a_{n+1} to be any element of $\bigcap_{m \leq n} X_m \setminus [s(n)]$ and notice that the resulting set $\{a_n \mid n \in \omega\}$ is indeed an unbounded pseudo-intersection of $(X_n)_{n \in \omega}$. $\square$

Concluding the section on small structures, it should be remarked that all of the previous proofs can be directly generalized to robustly d -Ramsey ultrafilters because they make no assumptions about the language L beside its countability, hence, any small structure can without loss of generality be expanded by a unary relation symbol interpreted by a cofinal set.

Remark 3.2.9. *All results regarding d -Ramsey ultrafilters in small ordered structures can be generalized to robustly d -Ramsey ultrafilters.*

Proof. This is clear because for a given small ordered structure $\mathfrak{A}$ and a cofinal set S the $L \cup \{C\}$ -structure $(\mathfrak{A}, S)$ for a new unary relation symbol C is still small and interprets a countable language. Hence, proving the existence of some d -Ramsey ultrafilter in arbitrary small structures implies its existence in the expansion $(\mathfrak{A}, S)$, yielding a robustly d -Ramsey ultrafilter in $\mathfrak{A}$. $\square$

3.3 Small Parameter Sets

The setting of small ordered structures can be further generalized when observing that the single benefit of a small universe is the small number of definable colorings, which can also be achieved by directly restricting the parameter set while allowing an arbitrary cardinality of the universe. This yields the following weaker ultrafilter properties.

Definition 3.3.1. Let $X \subseteq A$ be a parameter set. An ultrafilter is d_X -Ramsey if it contains an unbounded homogeneous set for every binary coloring definable over X . It is d_X -selective if for every partition $\mathcal{P}$ of A definable over X either $\mathcal{U} \cap \mathcal{P} \neq \emptyset$ or there exists an $X \in \mathcal{U}$ such that $|X \cap P| \leq 1$ for $P \in \mathcal{P}$.

The smallness of such a parameter set can now be defined analogously to small ordered structures, although the condition that $\text{cf}(\mathfrak{A})$ is countable is not required anymore. By Proposition 3.2.5 however, the following definition is equivalent to the parameter set being countable when $\text{cf}(\mathfrak{A})$ is uncountable.

Definition 3.3.2. A parameter set $X \subseteq A$ is *small* if $|X| \leq \mathfrak{p}_{\text{cf}(\mathfrak{A})}$.

Making no assumptions about the ordered structure and, in particular, not requiring that the order $<^{\mathfrak{A}}$ admits homogeneous sets, limiting the size of the parameter set is of course not sufficient for proving the existence of d_X -Ramsey ultrafilters. On the other hand, the next proposition shows that it implies the existence of d_X -selective ultrafilters, the crucial difference being that homogeneous sets for partitions exist regardless of the order $<^{\mathfrak{A}}$. This proof is analogous to the proof of Theorem 3.2.4.

Proposition 3.3.3. Let $X \subseteq A$ be small. Then, there exists a d_X -selective ultrafilter.

Proof. Let $\kappa := |X| \leq \mathfrak{p}_{\text{cf}(\mathfrak{A})}$, $s : \text{cf}(\mathfrak{A}) \rightarrow A$ an unbounded, increasing function, and let $\{\mathcal{P}_\alpha \mid \alpha \in \kappa\}$ be the set of all partitions definable over X . Choose for $X_0 \subseteq \text{im}(s)$ a homogeneous set for the partition of $\text{im}(s)$ induced by $\mathcal{P}_0$. Explicitly, if there exists an unbounded piece P , choose $X_0 := P$. By upwards-closedness, the constructed ultrafilter will then contain the corresponding piece of $\mathcal{P}_0$. Otherwise, every piece of the partition is bounded, so there are $\text{cf}(\mathfrak{A})$ many since choosing one bound for each piece yields an unbounded set, which is clear by the fact that $\mathcal{P}_0$ is a partition. The elements of the set X_0 , written as $\{a_\alpha \mid \alpha \in \text{cf}(\mathfrak{A})\}$, can now be inductively defined by choosing for a_0 the element $s(0)$ and for $a_{\alpha+1}$ the maximum of $s(\alpha + 1)$ and some bound $b_\alpha \in \text{im}(s)$ on the piece containing a_α . For a limit $\lambda \in \text{cf}(\mathfrak{A})$, select for a_λ the supremum of $\{s(\lambda)\} \cup \{b_\alpha \mid \alpha \in \lambda\}$. The constructed set is clearly cofinal, i. e., unbounded and of order type $\text{cf}(\mathfrak{A})$. Furthermore, it intersects with each piece of $\mathcal{P}_0$ in at most one element. This follows from the fact that choosing some a_α in a piece $P \in \mathcal{P}_0$ leads to all subsequent elements being chosen above the bound b_α and thus outside of P . Now assume that X_α with $\alpha \in \kappa$ is defined to be cofinal. Then, choose for $X_{\alpha+1}$ an unbounded homogeneous set for the partition of X_α induced by $\mathcal{P}_{\alpha+1}$ as before. If $\lambda \in \kappa$ is a limit ordinal, observe that $\mathcal{X}_\lambda := \{X_\alpha \mid \alpha < \lambda\}$ is a family of infinite subsets of $\text{im}(s)$ and it holds that $|\mathcal{X}_\lambda| < \kappa \leq \mathfrak{p}_{\text{cf}(\mathfrak{A})}$. Hence, since $\mathcal{X}_\lambda$ has the unbounded finite intersection property, it has an unbounded pseudo-intersection Y_λ , and $X_\lambda \subseteq Y_\lambda$ can be chosen as before. Then, the family $\{X_\alpha \mid \alpha \in \kappa\}$ has the unbounded finite intersection property and is therefore contained in an unbounded ultrafilter $\mathcal{U}$, which is d_X -selective by construction. $\square$

By the following corollary, the d_X -Ramsey property and d_X -selectivity can be properly separated in ordered structures that do not admit homogeneous sets for binary

colorings. Moreover, such structures exist by Corollary 3.4.6, proved in the next section.

Corollary 3.3.4. *Let $\mathfrak{A}$ be an ordered structure with a binary coloring definable over $X \subseteq A$ that has no unbounded homogeneous set. Then, there exists a d_X -selective ultrafilter that is not d_X -Ramsey.*

Proof. By Proposition 3.3.3, there exists a d_X -selective ultrafilter $\mathcal{U}$. By assumption there exists a coloring $\pi : [A]^2 \rightarrow 2$ without unbounded homogeneous sets, so $\mathcal{U}$ trivially does not contain such a set. $\square$

3.4 Uncountable Ordered Structures

This section comprises some considerations regarding unbounded homogeneous sets in ordered structures of arbitrary size. An invaluable tool in this context is the concept of cofinality, as could be seen in the previous sections, where it was required to be countable. It is therefore natural to ask what can be proved when assuming different cardinal properties for the cofinality. We borrow the following notation from Ramsey theory, which builds upon and generalizes Ramsey's theorem.

Definition 3.4.1. Let κ, λ be infinite cardinals, n a natural number, and μ a cardinal. Writing

$$\kappa \rightarrow (\lambda)_\mu^n$$

means that every coloring $\pi : [\kappa]^n \rightarrow \mu$ has a homogeneous set of size λ .

Using this notation, one can directly state a large cardinal property that, when assumed for $\text{cf}(\mathfrak{A})$, entails that $<^\mathfrak{A}$ admits homogeneous sets. This for example allows to strengthen Proposition 3.3.3 to the existence of d_X -Ramsey ultrafilters for every small parameter set X due to the existence of homogeneous sets for all colorings.

Definition 3.4.2. A cardinal κ is *weakly compact* if it is uncountable and satisfies $\kappa \rightarrow (\kappa)_2^2$.

This is the common definition, but it can be proved to be equivalent to the following, seemingly stronger property, which provides homogeneous sets for all n -ary colorings with $n \in \omega$.

Remark 3.4.3. *A cardinal κ is weakly compact if and only if it holds that $\kappa \rightarrow (\kappa)_\mu^n$ for all $n \in \omega, \mu \in \kappa$. [9, Chapter 2]*

Lemma 3.4.4. *If $\text{cf}(\mathfrak{A})$ is weakly compact, then every binary coloring of $\mathfrak{A}$ has an unbounded homogeneous set $X \subseteq A$.*

Proof. The proof is analogous to that of Lemma 3.1.2 and uses the fact that, by definition of weakly compact cardinals, there exists for each coloring $\pi : [\kappa]^2 \rightarrow 2$ for $\kappa := \text{cf}(\mathfrak{A})$ a homogeneous set of size κ . $\square$

Clearly, having homogeneous sets of size $\text{cf}(\mathfrak{A})$ for every binary coloring of $\text{cf}(\mathfrak{A})$ implies the existence of such sets for binary colorings of $\kappa := |A|$. On the other hand, assuming that $\mathfrak{A}$ does not even satisfy this weaker property yields that $<^{\mathfrak{A}}$ does not admit homogeneous sets, which in turn implies the existence of an expansion of $\mathfrak{A}$ admitting no d -Ramsey ultrafilters.

Proposition 3.4.5. *Let $\kappa := |\mathfrak{A}|$. If $\kappa \nrightarrow (\text{cf}(\mathfrak{A}))_2^2$, then there is a $\pi : [A]^2 \rightarrow 2$ that does not have an unbounded homogeneous set.*

Proof. By assumption there exists a $\pi : [A]^2 \rightarrow 2$ that does not have homogeneous sets of size $\text{cf}(\mathfrak{A})$, so all homogeneous sets are bounded. $\square$

Constructing an ordered structure without d -Ramsey ultrafilters can for example be achieved by utilizing κ -like structures for some regular but not weakly compact κ . Viable choices are successor cardinals or sufficiently small inaccessible cardinals. By κ -likeness, it follows that $\text{cf}(\mathfrak{A}) = \kappa$ and thus $\kappa \nrightarrow (\text{cf}(\mathfrak{A}))_2^2$. By the above proposition, there exists a coloring with no unbounded homogeneous set, so making this coloring definable by going to an appropriate expansion prohibits d -Ramsey ultrafilters. An analogous approach can for example be conducted using an ordered structure with $|A| = 2^\lambda$ and $\text{cf}(\mathfrak{A}) = \lambda^+$ for any cardinal λ since it is known that $2^\lambda \nrightarrow (\lambda^+)_2^2$. [7, Chapter 9]

Corollary 3.4.6. *Let κ be regular but not weakly compact. Then, there exists a κ -like ordered structure with a definable coloring that has no unbounded homogeneous set. Moreover, this structure has no d -Ramsey ultrafilters.*

While assuming a weakly compact cofinality implies the existence of unbounded homogeneous sets for all colorings, it appears to be insufficient for proving that they can be collected in an ultrafilter, even when restricting such a collection to definable colorings. However, this can be achieved with the notion of measurability, which constitutes a proper strengthening of weak compactness and implies the existence of Ramsey ultrafilters.

Definition 3.4.7. A cardinal κ is *measurable* if it is uncountable and there exists a κ -closed ultrafilter on κ .

The reason for the existence of Ramsey ultrafilters over measurable cardinals is the possibility to construct from a κ -closed ultrafilter another κ -closed ultrafilter that additionally is normal, defined below. As is shown in Proposition 3.4.10, such ultrafilters are Ramsey. This in turn implies the existence of Ramsey ultrafilters for every ordered structure $\mathfrak{A}$ with measurable cofinality.

Definition 3.4.8. For a family $\{X_\alpha \subseteq \kappa \mid \alpha \in \kappa\}$, the *diagonal intersection* is defined as the set $\Delta_{\alpha \in \kappa} X_\alpha := \{\alpha \in \kappa \mid \alpha \in \bigcap_{\beta \in \alpha} X_\beta\}$. An ultrafilter $\mathcal{U}$ over κ is *normal* if it is closed under diagonal intersections, i. e., if $X_\alpha \in \mathcal{U}$ for all $\alpha \in \kappa$, then $\Delta_{\alpha \in \kappa} X_\alpha \in \mathcal{U}$.

Lemma 3.4.9. *Every measurable cardinal has a normal, κ -closed ultrafilter. [7, Chapter 10]*

The following modifies the version of the proof of Ramsey's Theorem found in [7, Chapter 9] in such a way that it does not only show the existence of large homogeneous set for all binary colorings over κ but also their containedness in a fixed normal, κ -closed ultrafilter, which is therefore Ramsey. Moreover, this implies that every measurable cardinal is weakly compact.

Proposition 3.4.10. *Every measurable cardinal κ has a Ramsey ultrafilter.*

Proof. Let $\mathcal{U}$ be a normal, κ -complete ultrafilter over κ . We claim that $\mathcal{U}$ is Ramsey. Let $f : [\kappa]^2 \rightarrow 2$ be a coloring. Define $f_\alpha : \kappa \setminus \{\alpha\} \rightarrow 2$, $X \mapsto f(X \cup \{\alpha\})$ for each $\alpha \in \kappa$, and define $X_0 := \kappa$. Inductively choose for $X_{\alpha+1}$ a homogeneous set of cardinality κ for $f_\alpha \upharpoonright X_\alpha$ that is contained in $\mathcal{U}$. For a limit ordinal λ , define $X_\lambda := \bigcap_{\alpha \in \lambda} X_\alpha$ and observe that $X_\lambda \in \mathcal{U}$ by κ -completeness. Then, $X := \Delta_{\alpha \in \kappa} X_\alpha$ is in $\mathcal{U}$ by the normality of $\mathcal{U}$. Now for $\alpha \in X$, the coloring f_α is constant on the set $X \setminus (\alpha + 1)$ since it is a subset of X_α by definition of the diagonal intersection. Define the unary coloring $g : \kappa \rightarrow 2$ mapping α to this constant value of f_α . Since $\mathcal{U}$ is an ultrafilter, it contains a homogeneous set Y of cardinality κ for g . Denote its constant value by c . Then, Y is also homogeneous for f since for $\alpha, \beta \in Y$ with $\alpha < \beta$ it holds that $f(\{\alpha, \beta\}) = f_\alpha(\{\beta\}) = g(\alpha) = c$. $\square$

Corollary 3.4.11. *Let κ be a measurable cardinal. Then, every ordered structure with $\text{cf}(\mathfrak{A}) = \kappa$ has a Ramsey ultrafilter.*

Proof. Let $X \subseteq A$ be a cofinal set and $\mathcal{U}$ a Ramsey ultrafilter over X . Then, since every coloring $\pi : [A]^2 \rightarrow 2$ has an unbounded homogeneous set in $\mathcal{U}$, every extension of $\mathcal{U}$ to an ultrafilter over $\mathfrak{A}$ is Ramsey for $\mathfrak{A}$. $\square$

However, it should be remarked that it is unknown whether measurable cardinals are consistent with ZFC, and they are known to be inconsistent with $\text{ZFC} + \mathbf{V} = \mathbf{L}$, hence, their existence is a rather strong set-theoretic hypothesis. [9, Chapter 1] This is especially important when assuming an ultrafilter to be $\text{cf}(\mathfrak{A})$ -closed and cofinal, as has been done several times in Chapter 2, which implies that $\text{cf}(\mathfrak{A})$ is measurable.

Proposition 3.4.12. *If $\text{cf}(\mathfrak{A})$ is uncountable and there exists a $\text{cf}(\mathfrak{A})$ -closed, cofinal ultrafilter $\mathcal{U}$, then $\text{cf}(\mathfrak{A})$ is measurable.*

Proof. Let $s : \text{cf}(\mathfrak{A}) \rightarrow A$ be unbounded and increasing. The family $\{s^{-1}(X) \mid X \in \mathcal{U}, X \subseteq \text{im}(s)\}$ is clearly a $\text{cf}(\mathfrak{A})$ -complete ultrafilter on $\text{cf}(\mathfrak{A})$, hence, $\text{cf}(\mathfrak{A})$ is measurable. $\square$

The next results utilize the Löwenheim-Skolem theorem to prove the existence of ordered structures with d -Ramsey ultrafilters for every cardinality. The proof of Theorem 3.4.13 deviates from the chain construction used previously and instead shows the truth of sentences amounting to the existence of a d -Ramsey family with the unbounded finite intersection property in certain elementary extensions. As opposed to the previous proofs, it does not only use the fact that there is only a small number of definable colorings but for the first time requires the definability itself.

Theorem 3.4.13. *Let $\mathfrak{A}$ be an ordered structure with a d -Ramsey ultrafilter $\mathcal{U}$. Then, for every cardinality κ*

- *if $\kappa < |\mathfrak{A}|$, then there exists an elementary substructure with a d -Ramsey ultrafilter,*
or
- *if $\kappa > |\mathfrak{A}|$, then there exists an elementary extension with a d -Ramsey ultrafilter.*

Proof. Add to $\mathfrak{A}$ as a second sort the set system $\mathcal{U}_0 \subseteq \mathcal{U}$ with $|\mathcal{U}_0| = |A|$ choosing an unbounded homogeneous set for every definable coloring. Further, expand this two-sorted structure by a binary relation symbol interpreted as the usual $\in$ -relation. Write the resulting structure as $\hat{\mathfrak{A}} := (\mathfrak{A}, \mathcal{U}_0, \in)$. Then, $\hat{\mathfrak{A}}$ satisfies a set Φ of first order sentences stating that every set in $\mathcal{U}_0$ is unbounded, $\mathcal{U}_0$ has the finite intersection property, and $\mathcal{U}_0$ contains a homogeneous set for every definable coloring. By the Löwenheim-Skolem theorem, there exists a structure $\hat{\mathfrak{B}} := (\mathfrak{B}, \mathcal{V}_0, \in')$ with $|\mathfrak{B}| = |\mathcal{V}_0| = \kappa$ and $\hat{\mathfrak{B}} \preceq \hat{\mathfrak{A}}$ if $\kappa < |\mathfrak{A}|$ or $\hat{\mathfrak{B}} \succeq \hat{\mathfrak{A}}$ if $\kappa > |\mathfrak{A}|$. Without loss of generality assume that $\mathcal{V}_0$ consists of sets and $\in'$ is the usual $\in$ -relation, otherwise go to an isomorphic copy. Then, $\hat{\mathfrak{B}}$ satisfies Φ , implying that $\mathcal{V}_0$ can be extended to a d -Ramsey ultrafilter $\mathcal{V}$ over $\mathfrak{B}$. $\square$

Then, starting from an arbitrary countable structure, which has a d -Ramsey ultrafilter by Proposition 3.1.3, yields structures with d -Ramsey ultrafilters for every cardinality. Moreover, it is clear by the above proof that every elementary extension that can be constructed via this detour introducing the second sort has a d -Ramsey ultrafilter, including elementary extensions with special properties. This observation for example implies the existence of κ -saturated structures with d -Ramsey ultrafilters for every cardinal κ .

Corollary 3.4.14. *For every cardinality κ , there exists an ordered structure with a d -Ramsey ultrafilter.*

Proof. This follows from the above theorem and the fact that every countable ordered structure has a d -Ramsey ultrafilter. $\square$

Definition 3.4.15. A cardinal κ is *strongly inaccessible* if $2^\lambda < \kappa$ for all $\lambda < \kappa$.

The following result regarding the existence of saturated ordered structures with d -Ramsey ultrafilters can be proved as a corollary of Theorem 3.4.13 by first extending the given structure by the second sort as above and then using a common construction via iterative type realization and the above remark which observes the existence of κ -saturated elementary extensions with d -Ramsey ultrafilters. However, there is an alternative proof that is of independent interest because it follows the by now well-known chain construction and overcomes any limit step by, first, formulating the existence of the required set as a partial 1-type and, second, concluding its realization from the fact that the structure under investigation is saturated.

Proposition 3.4.16. *Let κ be strongly inaccessible. Then, there is a saturated ordered structure of cardinality κ that has a d -Ramsey ultrafilter.*

3 Existence

Proof as a corollary. Again, add to a countable structure $\mathfrak{A}$ the countable set system $\mathcal{U}_0$ as a second sort and expand it by a binary relation symbol interpreted as the usual $\in$ -relation, yielding the structure $\hat{\mathfrak{A}} := (\mathfrak{A}, \mathcal{U}_0, \in)$. Now, take a saturated elementary extension of size κ , which exists because κ is strongly inaccessible. Then, the second sort of this extension can be used to construct a d -Ramsey ultrafilter as before. $\square$

Proof via type realization. Add to a countable ordered structure $\mathfrak{A}$ its power set as a second sort and expand the resulting structure by a symbol interpreted as the usual $\in$ -relation, i. e., $\hat{\mathfrak{A}} := (\mathfrak{A}, \mathcal{P}(A), \in)$, take a countable elementary substructure $\hat{\mathfrak{A}}_0 \preceq \hat{\mathfrak{A}}$ making the second sort countable as well, and extend it to a saturated elementary extension $\hat{\mathfrak{B}} := (\mathfrak{B}, \mathcal{B}, \in') \succeq \hat{\mathfrak{A}}_0$ of cardinality κ . Without loss of generality, assume that the second sort $\mathcal{B}$ only consists of actual sets. Using only successor ordinals as indices, let $\varphi_1(x, y), \varphi_2(x, y), \dots$ enumerate all definitions of binary colorings, called $\pi_1, \pi_2, \dots$, and define $X_0 := B$. For $\alpha + 1 \in \kappa$, assume X_α is already defined to be an unbounded definable set. Then, define $X_{\alpha+1}$ to be an unbounded homogeneous set of $\pi \upharpoonright [X_\alpha]^2$ that is contained in the set system of $\hat{\mathfrak{B}}$, making it definable. Such a set exists because the sentence claiming the existence of such a set is true in $\hat{\mathfrak{A}}$ and thus in $\hat{\mathfrak{B}}$. For a limit ordinal $\lambda \in \kappa$, it remains to show that there is an unbounded definable set contained in $\bigcap_{\alpha < \lambda} X_\alpha$. Now, the set of formulas $\Phi(X) := \{\forall x(x \in X \rightarrow x \in X_\alpha) \mid \alpha < \lambda\} \cup \{\forall x \exists y(y \in X \wedge x < y)\}$ is consistent because any finite subset is realized by some X_α with $\alpha < \lambda$. Consequently, it is a partial 1-type describing an unbounded subset of $\bigcap_{\alpha < \lambda} X_\alpha$, which is realized by some Y due to saturation of $\hat{\mathfrak{B}}$. Choose $X_\lambda := Y$. Thus, the constructed non-increasing chain $(X_\alpha)_{\alpha < \kappa}$ has the unbounded finite intersection property and can be extended to an unbounded ultrafilter $\mathcal{U}$. By construction, $\mathcal{U}$ is a d -Ramsey ultrafilter of the saturated structure $\mathfrak{B}$. $\square$

Theorem 3.4.13 can be generalized to the existence of d -Ramsey ultrafilters extending certain families of sets. Contrary to the analogous remark in the setting of small ordered structures, Corollary 3.2.7, where the existence of an unbounded pseudo-intersection was sufficient, these families additionally have to be definable in the following sense.

Definition 3.4.17. A set Φ of L -formulas with at least one free variable *defines* the family $\Phi(\mathfrak{A}) := \{\varphi(\mathfrak{A}, \bar{b}) \mid \varphi(x, \bar{y}) \in \Phi, \bar{b} \in A\}$.

Then, if the family $\Phi(\mathfrak{A})$ can be extended to a d -Ramsey ultrafilter over the original structure $\mathfrak{A}$ and one chooses the set system $\mathcal{U}_0$ as an extension of $\Phi(\mathfrak{A})$, the 2-sorted elementary extension $\hat{\mathfrak{B}}$ satisfies countably many sentences expressing the fact that the second sort constitutes a d -Ramsey family with the unbounded finite intersection property extending the family $\Phi(\mathfrak{B})$, giving the desired d -Ramsey ultrafilter. This corollary might prove particularly helpful for establishing consistency results for arithmetic theories using ultrapower constructions.

Corollary 3.4.18. Let $\mathfrak{A}$ be an ordered structure and Φ a set of L -formulas such that $\Phi(\mathfrak{A})$ can be extended to a d -Ramsey ultrafilter over $\mathfrak{A}$. Then,

- for every cardinal $\kappa < |\mathfrak{A}|$ there exists an elementary substructure of $\mathfrak{A}$,

- for every cardinal $\kappa > |\mathfrak{A}|$ there exists an elementary extension of $\mathfrak{A}$,

denoted by $\mathfrak{B}$, such that $\Phi(\mathfrak{B})$ can be extended to a d -Ramsey ultrafilter over $\mathfrak{B}$.

Proof. This is a corollary of Theorem 3.4.13 and follows from modifying its proof by choosing the set system $\mathcal{U}_0$ as a superset of $\Phi(\mathfrak{A})$. $\square$

The last result proves the existence of ultrafilters satisfying the d - P -point property for a restricted, uniformly definable set of partitions under the assumption that the ordered structure $\mathfrak{A}$ satisfies a sufficiently strong fragment of arithmetic.

Definition 3.4.19. Let $\varphi(x, y, \bar{z})$ be an L -formula. An ultrafilter over $\mathfrak{A}$ is a d - P -point for φ if it contains a homogeneous set for each partition defined by $\varphi(x, y, \bar{a})$ for some $\bar{a} \in A$.

For the proof we require $\mathfrak{A}$ to satisfy induction and a form of sequence coding defined below. The idea is again to construct a non-increasing chain of sets witnessing the d - P -point property for all partitions in the uniformly definable family, while replacing the set-theoretic transfinite induction, which includes difficult to handle limit steps, by the arithmetic induction true in $\mathfrak{A}$, which only has successor steps. It is sufficient to find a formula that captures the inductive construction that is demonstrated for example in the proof of Proposition 3.1.6. Here, the sequence coding is used to talk about the choices made in the construction steps preceding the next one.

Definition 3.4.20. An ordered structure *enables sequence coding* if there is a formula $\text{seq}(x)$ and functions $\text{len}(x), \text{get}(x, y) \in L$ for which it satisfies

$$\begin{aligned} & \exists s(\text{seq}(s) \wedge \text{len}(s) = 0), \\ & \forall s \forall x \exists s' \exists l \forall l' < l (\text{seq}(s) \rightarrow (\text{seq}(s') \wedge \text{len}(s) = l \wedge \text{len}(s') = l + 1 \\ & \quad \wedge \text{get}(s', l) = x \wedge \text{get}(s, l') = \text{get}(s', l'))), \\ & \forall s \forall s' ((\text{len}(s) = \text{len}(s') \wedge \forall l < \text{len}(s) \text{get}(s, l) = \text{get}(s', l)) \rightarrow s = s'), \end{aligned}$$

stating, respectively, that there exists an empty sequence, that sequences can be extended by one element, and that each sequence has a unique code. Here, the formula $\text{seq}(x)$ defines the set of sequence codes, while $\text{len}(x)$ outputs the length of a given sequence x and $\text{get}(x, y)$ outputs the element of the sequence x at position y .

This ad hoc notion specifically tailored to inductive construction of sequences is easily justified since in the standard model of arithmetic $\mathbb{N}$ a sequence of numbers $n_0, \dots, n_l$, where n_i has the binary representation $b_{i,0} \dots b_{i,m_i}$, can for example be encoded by the number with binary representation

$$b_{0,0}b_{0,0} \dots b_{0,m_0}b_{0,m_0}01 \dots 01b_{l,0}b_{l,0} \dots b_{l,m_l}b_{l,m_l}.$$

Hence, for adequate choices of $\text{seq}(x)$, $\text{len}(x)$, and $\text{get}(x, y)$, the theory of true arithmetic proves sequence coding in the above sense. It should however be clear that much weaker arithmetic theories suffice.

Theorem 3.4.21. *Let $\varphi(x, y, \bar{z})$ be an L -formula. If $\mathfrak{A}$ satisfies induction and enables sequence coding, then it has a d - P -point for φ .*

Proof. We can assume without loss of generality that $\varphi(x, y, z)$ has exactly three free variables due to sequence coding. The goal is to construct a non-increasing chain $(X_a)_{a \in A}$ of unbounded sets such that for each $a \in A$ the set X_a witnesses the d - P -point property for the partition $\mathcal{P}_a$ defined by the $L(A)$ -formula $\varphi(x, y, a)$, i. e., it either has bounded intersection with each piece or is a subset of one. For convenience we define $X_{-1} := A$. We will now find a formula stating that an element $s \in A$ is a sequence encoding the inductive construction up until inclusively step $z \in A$. This means that s is a sequence of length $z+1$ and at each position $z' \leq z$ encodes either 0 if $X_{z'-1}$ already had bounded intersection with each piece of $\mathcal{P}_{z'}$, in which case $X_{z'-1}$ is inherited as $X_{z'}$, or $p+1$ for the minimal $p \in A$ such that $X_{z'-1} \cap \varphi(\mathfrak{A}, p, z')$, taken as $X_{z'}$, is unbounded. Observe that such a minimum exists because $\mathfrak{A}$ satisfies induction. This can be formalized as the formula

$$\begin{aligned} \sigma(s, z) := & \text{seq}(s) \wedge \text{len}(s) = z + 1 \\ & \wedge \forall b \exists a (b < a \wedge \chi(a, s, z + 1)) \\ & \wedge \forall z' \leq z ((\text{get}(s, z') = 0 \leftrightarrow \forall p \beta(p, s, z')) \\ & \wedge \forall z' \leq z \forall p ((\text{get}(s, z') = p + 1 \rightarrow \neg \beta(p, s, z')) \\ & \wedge \forall z' \leq z \forall p \forall p' ((\text{get}(s, z') = p + 1 \wedge p' < p) \rightarrow \beta(p', s, z'))), \end{aligned}$$

where

$$\chi(x, s, z) := \forall z' < z (\text{get}(s, z') = 0 \vee \exists p (\text{get}(s, z') = p + 1 \wedge \varphi(x, p, z')))$$

defines exactly the set X_{z-1} resulting from the first z construction steps specified in the sequence s and

$$\beta(p, s, z) := \exists b \forall a ((\varphi(a, p, z) \wedge \chi(a, s, z)) \rightarrow a < b)$$

states that $X_{z-1} \cap \varphi(\mathfrak{A}, p, z)$ is bounded. From top to bottom, the lines in the definition of $\sigma(s, z)$ require that s is a sequence of length $z+1$, that the set X_z resulting from the construction encoded by s is unbounded, and that the encoded construction steps conform to the description above. It is easy to verify that

$$\mathfrak{A} \models \forall z \exists! s \sigma(s, z)$$

and that these sequences are compatible in the obvious sense. Indeed, it is clear that there exists a unique such sequence for $z = 0$ and that a suitable sequence for z can be uniquely extended by one element to a sequence for $z+1$. Hence, applying induction yields unique sequences for all z . Denote by s_a for $a \in A$ the sequence such that $\mathfrak{A} \models \sigma(s_a, a)$. Choosing $X_a := \chi(\mathfrak{A}, s_a, a+1)$ for all $a \in A$ gives the desired non-increasing chain that can be extended to a d - P -point for $\varphi(x, y, z)$ over $\mathfrak{A}$. $\square$

As a corollary, the result can be generalized to an arbitrary finite number of formulas by formalizing, similarly as above, a construction that alternates between construction steps for the different formulas. A precise proof is omitted because it is a complication of the previous one while providing no further conceptual insight.

Corollary 3.4.22. *Let $\varphi_1(x, y, \bar{z}), \dots, \varphi_n(x, y, \bar{z})$ with $n \in \omega$ be L -formulas. If $\mathfrak{A}$ satisfies induction and enables sequence coding, then it has a d - P -point for $\varphi_1, \dots, \varphi_n$.*

Furthermore, it is possible to find for each $n \in \omega$ a universal Σ_n -formula that, for an adequate parameter, is equivalent to any given Σ_n -formula, assuming that $\mathfrak{A}$ satisfies a sufficiently strong fragment of arithmetic. The proof is closely related to the universal Turing machine, which is capable of simulating the computation of any other Turing machine when given its code as an input. Applying Theorem 3.4.21 to this universal formula then yields a d - P -point for all Σ_n -formulas. See for example [10, Chapter 6] for a related result in bounded arithmetic, namely the existence of a S_2^1 -provable universal Σ_n^b -formula.

3.5 Separation

This section discusses the matter of existence of ultrafilters that are explicitly not d -Ramsey but still have weaker combinatorial properties. The goal is to achieve a better understanding of how these properties relate to each other. It is natural to ask whether the d - P -point property or d -selectivity is strictly weaker than being d -Ramsey. However, proving the existence of an ultrafilter with such a property while showing that it does not contain any homogeneous set for a fixed definable coloring appears difficult. It seems to be a promising approach to take an $\aleph_1$ -like ordered structure with a d - P -point and then find a coloring with no unbounded homogeneous sets, disproving the existence of Ramsey ultrafilters. The difficulty lies in showing that such a coloring is definable in order to refute d -Ramsey ultrafilters as well. Aside from such an approach, one could also directly show that the set of complements of unbounded homogeneous sets for a fixed definable coloring together with carefully chosen sets witnessing the d - P -point property have the unbounded finite intersection property. Retreating to consistency results, such a proof can be executed when assuming a weak form of Martin's Axiom because this implies the existence of a P -point over ω which is not Ramsey. [6, Chapter 14] This can be used to construct an ordered structure conforming to the context of this thesis over which a d - P -point exists that is not d -Ramsey.

Proposition 3.5.1. *It is consistent with ZFC that there exists an ordered structure such that there is a d - P -point which is not d -Ramsey.*

Proof. This is a corollary of a related result stating that the existence of a P -point over ω which is not Ramsey is consistent with ZFC. Assuming that there is such a P -point $\mathcal{U}$, the desired structure can be constructed by taking $\mathfrak{A} := (\omega, \in)$ and expanding it by some binary relation symbol that is interpreted such that it defines some binary coloring with no infinite homogeneous set in $\mathcal{U}$. Then, $\mathcal{U}$ is a d - P -point, but not d -Ramsey. $\square$

The next result shows the equivalence of a certain cardinal property and the existence of an ordered structure of weakly compact size not admitting homogeneous sets over which a selective ultrafilter exists. Therefore, assuming there exists a cardinal satisfying the required conditions implies that Ramsey and selective ultrafilters are not equivalent in general in the non-standard setting. Making an appropriate coloring definable by going to an expansion then yields the analogous result for the definable setting. Should the cardinal property turn out to be inconsistent, however, then every ordered structure of weakly compact size with a selective ultrafilter admits unbounded homogeneous sets for all colorings.

Proposition 3.5.2. *For a weakly compact cardinal κ , there exists an ordered structure $\mathfrak{A}$ of size κ with a selective ultrafilter such that there is a coloring with no unbounded homogeneous set if and only if there exists a Ramsey ultrafilter $\mathcal{U}$ over κ and a partition $\mathcal{P} := \{p_\alpha \mid \alpha \in \lambda\}$ of κ into infinite parts with regular but not weakly compact $\lambda := |\mathcal{P}| < \kappa$ such that $\bigcap_{\beta < \alpha} \kappa \setminus p_\beta \in \mathcal{U}$ for all $\alpha \in \lambda$.*

Proof. For the forward direction, observe that, by the existence of a coloring with no unbounded homogeneous set, $\text{cf}(\mathfrak{A}) < \kappa$. Otherwise, by the weak compactness of κ , every coloring would have an unbounded homogeneous subset of S for a cofinal S . For an unbounded increasing function $s : \text{cf}(\mathfrak{A}) \rightarrow A$, define a partition of A by $p_\alpha := [s(\alpha)] \setminus \bigcup_{\beta < \alpha} [s(\beta)]$ for all $\alpha \in \text{cf}(\mathfrak{A})$. Without loss of generality, choose s such that these pieces are infinite. Now, take an arbitrary bijection between A and κ and consider both the imported ultrafilter and the partition. These have the desired properties.

For the backward direction, construct a structure as follows. First, define B_α to be any ordered structure modeling $T_<$ of size $|p_\alpha|$ for all $\alpha \in \lambda$. Now, the desired structure $\mathfrak{A}$ is chosen of order type $\sum_{\alpha < \lambda} B_\alpha$, which is clearly a model of $T_<$. Denote by S_α the segment corresponding to B_α . For a bijection between κ and $\mathfrak{A}$ mapping the elements of each p_α onto S_α , the image of $\mathcal{U}$ becomes an unbounded selective ultrafilter over $\mathfrak{A}$. Regarding the non-existence of unbounded homogeneous sets, take any coloring $\pi : [\lambda]^2 \rightarrow 2$ of λ without a homogeneous set of size λ and inspect the coloring $\pi' : [A]^2 \rightarrow 2$ mapping $\{a, b\}$ to 0 if the lie in the same segment S_α for some $\alpha \in \lambda$ and $\pi(\{\alpha, \beta\})$ if the lie in the segments corresponding to distinct p_α and p_β , respectively. If π' has an unbounded homogeneous set, then it also has an unbounded homogeneous set X that intersects with each segment of $\mathfrak{A}$ in at most one element since each segment is bounded. Then, the set $\{\alpha \in \lambda \mid |X \cap S_\alpha| = 1\}$ is an unbounded homogeneous set of π . By the regularity of λ , this set has size λ itself, contradicting the choice of π . $\square$

4 Restricted Ultrapowers

This chapter introduces the concept of restricted ultrapowers and demonstrates how they can be used to establish consistency results for the theory $\forall\text{PV}$ from bounded arithmetic. More specifically, Section 4.2.3 proves a slight generalization of the main theorem in [1], where an unbounded ultrafilter over a Herbrand-saturated model of $\forall\text{PV}$ is used to construct a model satisfying a certain circuit lower bound. See below for the details. Note, however, that the proof does not involve any of the combinatorial properties discussed in Chapter 2. Finding an application for those is a matter of work in progress.

4.1 Basics

In the following section, fix a countable language L containing the binary relation symbol $<$ and an ordered L -structure $\mathfrak{A}$ satisfying the theory $T_{<}$ from Definition 2.1.7. The central idea of ultrapowers over $\mathfrak{A}$ is to build a structure from a set of functions from a non-empty set Ω to A and identifying two functions if they are equal on a large set in the sense of an ultrafilter $\mathcal{U}$. The interpretation of L by $\mathfrak{A}$ can then naturally be extended to such a set, given that it is closed in an appropriate sense. Since we are primarily interested in definable functions, all of which have domain A , we require in the following that Ω is a subset of A .

Definition 4.1.1. For a non-empty $\Omega \subseteq A$, let A^Ω denote the set of functions from Ω to A . A set of functions $B \subseteq A^\Omega$ is *closed under f* for an r -ary $f : A^r \rightarrow A$ if for every $\bar{\alpha} = (\alpha_0, \dots, \alpha_{r-1}) \in B^r$ it also contains $f \circ \bar{\alpha}$. It is *L -closed* if it is non-empty and closed under $f^\mathfrak{A}$ for every $f \in L$.

For such an L -closed set of functions, the identification of similar functions is made precise by the following equivalence relation induced by a filter on Ω . The resulting set of equivalence classes is then the universe of the ultrapower.

Definition 4.1.2. For an L -closed $B \subseteq A^\Omega$ and a filter $\mathcal{F}$ over Ω , let $B/\mathcal{F}$ denote the set of equivalence classes $\alpha^\mathcal{F}$ for $\alpha \in B$ with respect to $\sim^\mathcal{F}$ where $\alpha \sim^\mathcal{F} \beta$ if and only if $\{\omega \in \Omega \mid \alpha(\omega) = \beta(\omega)\} \in \mathcal{F}$.

Definition 4.1.3. For an L -formula $\varphi(\bar{x})$ and functions $\bar{\alpha} \in A^\Omega$ denote by $\langle \varphi(\bar{\alpha}) \rangle$ the set $\{\omega \in \Omega \mid \mathfrak{A} \models \varphi(\bar{\alpha}(\omega))\}$. For an L -closed $B \subseteq A^\Omega$ and a filter $\mathcal{F}$ over Ω , the *restricted power* $B/\mathcal{F}$ is the L -structure with universe $B/\mathcal{F}$ interpreting r -ary relation symbols R and function symbols f , respectively, by

4 Restricted Ultrapowers

- $R^{B/\mathcal{F}} := \{\bar{\alpha}^{\mathcal{F}} \in (B/\mathcal{F})^r \mid \langle R\bar{\alpha} \rangle \in \mathcal{F}\}$
- the function mapping $\bar{\alpha}^{\mathcal{F}} \in (B/\mathcal{F})^r$ to $(f^{\mathfrak{A}} \circ \bar{\alpha}^{\mathcal{F}}) \in B/\mathcal{F}$

It is called *restricted ultrapower* if $\mathcal{F}$ is an ultrafilter.

As a demonstration of how the different components of the above definition interact, we prove the following proposition stating that, if an ultrapower is constructed over a set of definable functions over $\mathfrak{A}$, then it is already fully determined by the definable sets contained in the ultrafilter.

Proposition 4.1.4. *Let $\Omega = A$, let $\mathcal{U}$ be a d -ultrafilter,*

$$\hat{\mathcal{U}} := \{X \subseteq A \mid \text{there exists a } Y \in \mathcal{U} \text{ s. t. } Y \subseteq X\}$$

the filter generated by $\mathcal{U}$, and let $D \subseteq A^\Omega$ be an L -closed set of definable functions. Then, for every ultrafilter $\mathcal{V} \supseteq \mathcal{U}$ it holds that $D/\mathcal{V} = D/\hat{\mathcal{U}}$.

Proof. First, show that the equivalence relations defined by these filters are equal. For $\alpha, \beta \in D$ definable by $\varphi(x, y)$ and $\psi(x, y)$, $\alpha \sim^{\hat{\mathcal{U}}} \beta$ clearly implies $\alpha \sim^{\mathcal{V}} \beta$ since $\hat{\mathcal{U}} \subseteq \mathcal{V}$. The converse holds because the set $X := \{a \in A \mid \alpha(a) = \beta(a)\} \in \mathcal{V}$ is definable by the formula

$$\chi(x) := \exists y(\varphi(x, y) \wedge \psi(x, y))$$

and thus $X \in \mathcal{U} \subseteq \hat{\mathcal{U}}$ since $\mathcal{U}$ contains exactly the definable sets in $\mathcal{V}$. Consequently, both structures have the same universe. It remains to show that they interpret the symbols in L in the same way, which trivially holds for function symbols. For an r -ary relation symbol R , it holds that $R^{D/\hat{\mathcal{U}}} \subseteq R^{D/\mathcal{V}}$ by $\hat{\mathcal{U}} \subseteq \mathcal{V}$. For the converse, observe that for $\bar{\alpha} := \alpha_1, \dots, \alpha_r \in D$ definable by $\varphi_1(x, y), \dots, \varphi_r(x, y)$ the set $Y := \{a \in A \mid \bar{\alpha}(a) \in R^{\mathfrak{A}}\}$ is definable in $\mathfrak{A}$ by the formula

$$\rho(x) := \exists y_1 \dots \exists y_r \left(\bigwedge_{i \in \{1, \dots, r\}} \varphi_i(x, y_i) \wedge R y_1 \dots y_r \right).$$

Thus, if Y is contained in $\mathcal{V}$, it is also contained in $\hat{\mathcal{U}}$, yielding $R^{D/\mathcal{V}} \subseteq R^{D/\hat{\mathcal{U}}}$. $\square$

Now, it is known by Łoś's theorem that an ultrapower $\mathfrak{A}^\Omega/\mathcal{U}$ over the full set of functions from Ω to A is an elementary extension of $\mathfrak{A}$ and thus unsuitable for proving consistency results since any sentence is true in the ultrapower if and only if it is true in $\mathfrak{A}$. This motivates the investigation of restricted ultrapowers and which sentences they preserve and leads to the notion of Łoś formulas, which, intuitively, are true under a given assignment in the ultrapower if and only if they are true in $\mathfrak{A}$ on a $\mathcal{U}$ -big subset of Ω . In particular, the ultrapower preserves all Łoś sentences.

Definition 4.1.5. An L -formula $\varphi(\bar{x})$ is *Łoś* if

$$F/\mathcal{U} \models \varphi(\bar{\alpha}^{\mathcal{U}}) \iff \langle \varphi(\bar{\alpha}) \rangle \in \mathcal{U}$$

for every L -closed $F \subseteq A^\Omega$, every ultrafilter $\mathcal{U}$ over Ω , and all $\bar{\alpha} \in F$.

Since it is generally not clear how this set $\langle \varphi(\bar{\alpha}) \rangle$ looks like, it is worth a side note that, by Proposition 4.1.4, it has in fact a $\mathcal{U}$ -big definable subset in the case that $F \subseteq A^\Omega$ is a set of definable functions.

Corollary 4.1.6. *Let $F \subseteq A^\Omega$ be an L -closed set of definable functions, let $\mathcal{U}$ be an ultrafilter over Ω and write $\mathcal{S} = \mathcal{U} \cap D(\mathfrak{A})$. Further, let $\varphi(\bar{x})$ be Łoś and $\bar{\alpha} \in F$. Then, $F/\mathcal{U} \models \varphi(\bar{\alpha}^\mathcal{U})$ if and only if $\langle \varphi(\bar{\alpha}) \rangle \in \hat{\mathcal{S}}$ for $\hat{\mathcal{S}}$ the filter generated by $\mathcal{S}$.*

Proof. The backward direction is clear by definition. The forward direction follows from Proposition 4.1.4 implying that $F/\mathcal{U} = F/\mathcal{V}$ for all $\mathcal{V} \supseteq \mathcal{S}$ and thus, since $\varphi(\bar{x})$ is Łoś, $\langle \varphi(\bar{\alpha}) \rangle \in \mathcal{V}$ for all such $\mathcal{V}$. Then, $\langle \varphi(\bar{\alpha}) \rangle \in \bigcap_{\mathcal{V} \supseteq \mathcal{S}} \mathcal{V} = \hat{\mathcal{S}}$. For the last equality observe that, if $X \in \mathcal{W} \setminus \hat{\mathcal{S}}$ for $\mathcal{W} \supseteq \mathcal{S}$, the set $\hat{\mathcal{S}} \cup \{\Omega \setminus X\}$ has the finite intersection property. Otherwise, there exists a $Y \in \hat{\mathcal{S}}$ such that $(\Omega \setminus X) \cap Y = \emptyset$ by the filter properties of $\hat{\mathcal{S}}$, so $Y \subseteq X$ and therefore $X \in \hat{\mathcal{S}}$, contradicting the choice of X . Consequently, there exists an ultrafilter $\mathcal{W}' \supseteq \hat{\mathcal{S}} \cup \{\Omega \setminus X\}$ not containing X , so $X \notin \mathcal{W} \cap \mathcal{W}' \supseteq \bigcap_{\mathcal{V} \supseteq \mathcal{S}} \mathcal{V}$. $\square$

Returning to the general setting, the following two propositions show that, first, all quantifier-free formulas are Łoś and, second, the universal closure of a Łoś formula is true in the ultrapower if it is true in $\mathfrak{A}$.

Proposition 4.1.7. *Quantifier-free formulas are Łoś.*

Proof. Atomic formulas are Łoś by Definition 4.1.3, and Łoś formulas are closed under Boolean combinations. Let $F \subseteq A^\Omega$ be L -closed, $\mathcal{U}$ an ultrafilter over Ω , $\bar{\alpha} \in F$, and let $\varphi(\bar{x}), \psi(\bar{x})$ be Łoś. Then

$$F/\mathcal{U} \models \neg \varphi(\bar{\alpha}^\mathcal{U}) \iff F/\mathcal{U} \not\models \varphi(\bar{\alpha}^\mathcal{U}) \iff \langle \varphi(\bar{\alpha}) \rangle \notin \mathcal{U} \iff \langle \neg \varphi(\bar{\alpha}) \rangle \in \mathcal{U}$$

and

$$\begin{aligned} F/\mathcal{U} &\models \varphi(\bar{\alpha}^\mathcal{U}) \wedge \psi(\bar{\alpha}^\mathcal{U}) \\ &\iff F/\mathcal{U} \models \varphi(\bar{\alpha}^\mathcal{U}) \text{ and } F/\mathcal{U} \models \psi(\bar{\alpha}^\mathcal{U}) \\ &\iff \langle \varphi(\bar{\alpha}) \rangle \in \mathcal{U} \text{ and } \langle \psi(\bar{\alpha}) \rangle \in \mathcal{U} \\ &\iff \langle \varphi(\bar{\alpha}) \rangle \cap \langle \psi(\bar{\alpha}) \rangle \in \mathcal{U} \\ &\iff \langle \varphi(\bar{\alpha}) \wedge \psi(\bar{\alpha}) \rangle \in \mathcal{U} \end{aligned}$$

by the definitions of Łoś formulas, ultrafilters, and ultrapowers, verifying that Łoś formulas are closed under Boolean combinations. $\square$

Proposition 4.1.8. *For an L -closed $F \subseteq A^\Omega$ and an ultrafilter $\mathcal{U}$ over Ω , if an L -formula $\varphi(\bar{x})$ is Łoś and $\mathfrak{A} \models \forall \bar{x} \varphi(\bar{x})$, then $F/\mathcal{U} \models \forall \bar{x} \varphi(\bar{x})$.*

Proof. If $\mathfrak{A} \models \forall \bar{x} \varphi(\bar{x})$, then also $\mathfrak{A} \models \forall x \varphi(\bar{\alpha}(x))$ for all $\bar{\alpha} \in F$, so $\langle \varphi(\bar{\alpha}) \rangle = \Omega$. Since $\varphi(\bar{x})$ is Łoś and $\Omega \in \mathcal{U}$, this implies that $F/\mathcal{U} \models \varphi(\bar{\alpha}^\mathcal{U})$ for all $\bar{\alpha}$ and thus $F/\mathcal{U} \models \forall \bar{x} \varphi(\bar{x})$. $\square$

The last proposition can be further strengthened to the fact that the structure $\mathfrak{A}$ is isomorphic to an existentially closed substructure of any given restricted ultrapower over an $L(A)$ -closed set of functions. The most significant consequence of this is that, if $\mathfrak{A}$ models a fixed universal theory, then so does the ultrapower, making restricted ultrapowers a viable choice for model-theoretic proofs of consistency results.

Definition 4.1.9. For a structure $\mathfrak{A}$ a substructure $\mathfrak{B} \subseteq \mathfrak{A}$ is *existentially closed* if for all quantifier-free L -formulas $\varphi(\bar{x}, y)$ and parameters $\bar{b} \in B$ with $\mathfrak{A} \models \exists y \varphi(\bar{b}, y)$ also $\mathfrak{B} \models \exists y \varphi(\bar{b}, y)$.

Proposition 4.1.10. For an $L(A)$ -closed $F \subseteq A^\Omega$ and an ultrafilter $\mathcal{U}$ over Ω , the map $\pi : A \rightarrow F/\mathcal{U}, a \mapsto \alpha_a^\mathcal{U}$ is an isomorphism onto an existentially closed substructure of $F/\mathcal{U}$.

Proof. The map is clearly an isomorphism onto $\pi(A)$ since quantifier-free formulas are Łoś and thus preserved in $F/\mathcal{U}$. Moreover, $\pi(A)$ is existentially closed in $F/\mathcal{U}$. This can be concluded by considering $\mathfrak{A}$ and $F/\mathcal{U}$ as $L(A)$ -structures and applying Proposition 4.1.8. $\square$

4.2 Consistency of Circuit Lower Bounds

Having the tool of restricted ultrapowers at hand, we are going to prove the consistency of a circuit lower bound with the universal theory $\forall\text{PV}$ of bounded arithmetic. This section first gives a formal definition of the theory and the lower bound, continues with a brief excursion on the model-theoretic notion of Herbrand saturation, and then puts everything together for the consistency proof.

4.2.1 The Theory $\forall\text{PV}$

Definition 4.2.1. The language PV contains a binary relation symbol $<$ and r -ary function symbols for each polynomial time computable function $f : \mathbb{N}^r \rightarrow \mathbb{N}$ with $r \in \mathbb{N}$. The theory $\forall\text{PV}$ is the set of all universal sentences true in the standard model $\mathbb{N}$ of PV , which has as universe the natural numbers and interprets $<$ by their usual order and each function symbol by the corresponding function.

The goal is to show that for each $k \in \mathbb{N}$ there exists a function symbol $g(x) \in \text{PV}$ such that there exists a model of $\forall\text{PV}$ satisfying that each non-uniform size n^k circuit family for infinitely many input lengths fails to compute $g(x)$ correctly. For a formalization of circuit lower bounds in the theory $\forall\text{PV}$, a definition of circuits and how they are coded by numbers is required. We restrict our considerations to circuits built from $\wedge$ -, $\vee$ -, and $\neg$ -gates, each of which has fan-in at most 2, with exactly one output gate. Such a circuit C with s many gates, where s is called the *size* of C , is then coded by the number coding the set of triples $\langle u, v, w \rangle$, where $u, v < s$ are indices of gates such that the output of gate u is an input of gate v , and w codes the label $\wedge$, $\vee$, or $\neg$ of v . Further, $\text{Circuit}(x, y)$ is the quantifier-free formula that defines in the standard model $\mathbb{N}$ of PV

the set of pairs (C, s) where C is a circuit of size at most s , and $\text{eval}(x, y) \in \mathbf{PV}$ is the function mapping (C, a) to the output bit of C on the input a . We define $0 \in \mathbb{N}$ to be a circuit of size 0, i. e., $\text{Circuit}(0, 0) \in \forall\mathbf{PV}$. Circuit lower bounds can then be formalized in $\forall\mathbf{PV}$ as follows.

Definition 4.2.2. Define

$$\text{LB}_f(s, n) := n > 0 \wedge \forall C (\text{Circuit}(C, s) \rightarrow \exists x (|x| = n \wedge f(x) \neq \text{eval}(C, x)))$$

as the formula expressing that there is no circuit of size at most s computing $f(x)$ correctly on all inputs of length n .

In the following, fix a model $\mathfrak{A}$ of $\forall\mathbf{PV}$. Let $|x| = \lceil \log_2(x+1) \rceil$ denote the function that in the standard model maps a number x to the length of its binary representation, which is clearly in $\mathbf{PV}$. We write $\text{Len}(\mathfrak{A})$ for the set $\{a \in A \mid \mathfrak{A} \models \exists b \ a = |b|\}$. This allows us to state the theorem in the following formally precise way.

Theorem 4.2.3. *For every $k \in \mathbb{N}$ there exists a Boolean function $g(x) \in \mathbf{PV}$ such that $\forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_g(|z| \cdot l^k, l))$ is consistent with $\forall\mathbf{PV}$.*

The model of $\forall\mathbf{PV}$ witnessing the lower bound consistency will be a restricted ultrapower over the set of polynomial time computable functions in the following sense.

Definition 4.2.4. We define $\mathbf{PTIME}(\mathfrak{A}) := \{t^{\mathfrak{A}} \mid t(x) \text{ is a } \mathbf{PV}(A)\text{-term}\}$. A *polynomial time ultrapower* is the structure $\mathbf{PTIME}(\mathfrak{A})/\mathcal{U}$ for an ultrafilter $\mathcal{U}$ over $\mathfrak{A}$.

For simplicity, however, we are going to restrict the function domain Ω to the set $\{1^n \mid n \in \text{Len}(\mathfrak{A})\}$, where 1^n is $(1 \# a) - 1$ for $n = |a|$ and $\# \in \mathbf{PV}$ the *smash* function with $n \# m := 2^{|n| \cdot |m|}$. This is the indeed without loss of generality by the following remark.

Remark 4.2.5. *If $\mathcal{U}$ is an ultrafilter over Ω and $\mathcal{V} \supseteq \mathcal{U}$ is an ultrafilter over A , then the map $(\alpha \restriction \Omega)^{\mathcal{U}} \mapsto \alpha^{\mathcal{V}}$ is an isomorphism from $(\mathbf{PTIME}(\mathfrak{A}) \restriction \Omega)/\mathcal{U}$ onto $\mathbf{PTIME}(\mathfrak{A})/\mathcal{V}$. [11]*

Another useful property of $\forall\mathbf{PV}$ is that it supports definition by cases. As a trivial but convenient consequence, every $\mathbf{PV}$ -term is $\forall\mathbf{PV}$ -provably equivalent to a single function symbol in $\mathbf{PV}$.

Definition 4.2.6. An L -theory T *supports definition by cases* if for all $k \in \mathbb{N}$, all quantifier-free L -formulas $\varphi_0(\bar{x}), \dots, \varphi_{k-1}(\bar{x})$, all L -terms $t_0(\bar{x}), \dots, t_k(\bar{x})$ there exists a function symbol $f \in L$ such that for all $0 \leq j < k$

$$\begin{aligned} T \vdash \bigwedge_{i=0}^{j-1} \neg \varphi_i(\bar{x}) \wedge \varphi_j(\bar{x}) &\rightarrow f(\bar{x}) = t_j(\bar{x}), \\ T \vdash \bigwedge_{i=0}^{k-1} \neg \varphi_i(\bar{x}) &\rightarrow f(\bar{x}) = t_k(\bar{x}). \end{aligned}$$

Remark 4.2.7. $\forall\text{PV}$ supports definition by cases.

Proof. For all $k \in \mathbb{N}$, quantifier-free PV-formulas $\varphi_0(\bar{x}), \dots, \varphi_{k-1}(\bar{x})$, and PV-terms $t_0(\bar{x}), \dots, t_k(\bar{x})$, the desired function can be computed in polynomial time in the standard model by a Turing machine that successively computes the truth values of the formulas and outputs the value of the correct term accordingly. The defining equations of this function are then universal and thus axioms of $\forall\text{PV}$. $\square$

This, combined with sequence coding, leads to the observation that $\text{PTIME}(\mathfrak{A}) = \{f_a^\mathfrak{A} \mid f(x, y) \in \text{PV}, a \in A\}$, where $f_a^\mathfrak{A}(x)$ abbreviates $f^\mathfrak{A}(x, a)$.

Definition 4.2.8. Let δ be rational, $\mathfrak{A}$ a model of $\forall\text{PV}$. A function $f_a^\mathfrak{A} \in \text{PTIME}(\mathfrak{A})$ is a *uniform size n^δ circuit family* in $\mathfrak{A}$ if for all $n \in \text{Len}(\mathfrak{A})$

$$\mathfrak{A} \models \text{Circuit}(f_a(1^n), n^\delta).$$

4.2.2 Herbrand Saturation

This subsection is based on [12]. The notion of Herbrand saturation, which will play a key role in the proof of Theorem 4.2.3, can be understood as the semantic counterpart of the following syntactic witnessing theorem.

Theorem 4.2.9 (Herbrand). *Let T be a universal L -theory and $\varphi(\bar{x}, \bar{y})$ a quantifier-free L -formula such that $T \vdash \exists \bar{y} \varphi(\bar{x}, \bar{y})$. Then, there exist sequences of L -terms $\bar{t}_0(\bar{x}), \dots, \bar{t}_k(\bar{x})$ such that $T \vdash \bigvee_{i=0}^k \varphi(\bar{x}, \bar{t}_i(\bar{x}))$.*

Proof. Otherwise, T is by compactness consistent with the set

$$S := \{\neg\varphi(\bar{c}, \bar{t}(\bar{c})) \mid \bar{t}(\bar{x}) \text{ is a sequence of } L\text{-terms}\}$$

where $\bar{c}$ is a sequence of new constants. Thus, $T \cup S$ has a model $\mathfrak{A}$. The submodel $\mathfrak{B} \subseteq \mathfrak{A}$ generated by the set of terms $\{t(\bar{c}) \mid t \text{ is an } L\text{-term}\}$ is again a model of $T \cup S$ since this theory is universal, and by construction $\mathfrak{B} \not\models \exists \bar{y} \varphi(\bar{c}, \bar{y})$. $\square$

Herbrand saturation can then be intuitively described as replacing provability in Herbrand's theorem by truth.

Definition 4.2.10. An L -structure $\mathfrak{A}$ is *Herbrand-saturated* if for every quantifier-free L -formula $\varphi(\bar{x}, \bar{y}, \bar{z})$ and parameters $\bar{a} \in \mathfrak{A}$ with $\mathfrak{A} \models \forall \bar{x} \exists \bar{y} \varphi(\bar{x}, \bar{y}, \bar{a})$ there exist parameters $\bar{b} \in \mathfrak{A}$ and sequences of terms $\bar{t}_0(\bar{x}, \bar{b}), \dots, \bar{t}_k(\bar{x}, \bar{b})$ such that

$$\mathfrak{A} \models \forall \bar{x} \bigvee_{i=0}^k \varphi(\bar{x}, \bar{t}_i(\bar{x}, \bar{b}), \bar{a}).$$

Even though this property appears to be rather strong, Herbrand-saturated models can be proved to exist for any consistent universal theory. In particular, they exist for $\forall\text{PV}$.

Theorem 4.2.11. *Every consistent universal theory has a Herbrand-saturated model.*

Proof. Let T be a consistent universal L -theory for a countable language L . The argument can easily be generalized to uncountable settings by using a transfinite iteration in the definition below. Define $L_\omega := L \cup \{c_n \mid n \in \omega\}$ to be L with countably many new constant symbols and consider the enumeration $\varphi_1(\bar{x}_1, \bar{y}_1), \varphi_2(\bar{x}_2, \bar{y}_2), \dots$ of all L_ω -formulas with all possible partitions of their free variables. Now, define an ω -chain of consistent universal theories

$$S_0 := T,$$

$$S_{i+1} := \begin{cases} S_i \cup \{\forall \bar{y}_{i+1} \varphi_{i+1}(\bar{c}, \bar{y}_{i+1})\} & \text{if consistent} \\ S_i & \text{otherwise,} \end{cases}$$

where in step $i + 1$ the constants $\bar{c}$ are chosen such that they neither occur in S_i nor in $\varphi_{i+1}(\bar{x}_{i+1}, \bar{y}_{i+1})$. By compactness, the theory $S_\omega := \bigcup_{i \in \omega} S_i$ is consistent. For a model $\mathfrak{B}$ of S_ω , take the substructure $\mathfrak{A}$ generated by closed L_ω -terms, which is again a model of S_ω and consequently of T because S_ω is universal. Notice that every element of $\mathfrak{A}$ generated by an L_ω -term t is also generated by some constant because the L_ω -formula $t = x$ is enumerated at some point in the construction of the chain.

It can now be shown that $\mathfrak{A} \upharpoonright L$ is a Herbrand-saturated model of T . Assume $\mathfrak{A} \upharpoonright L \models \forall \bar{x} \exists \bar{y} \psi(\bar{x}, \bar{y}, \bar{a})$ for a quantifier-free L -formula $\psi(\bar{x}, \bar{y}, \bar{z})$ and parameters $\bar{a} \in \mathfrak{A}$. Then, clearly $\mathfrak{A} \not\models \exists \bar{x} \forall \bar{y} \neg \psi(\bar{x}, \bar{y}, \bar{a})$. Take constants $\bar{d}$ such that $\bar{d}^{\mathfrak{A}} = \bar{a}$ and consider the construction step $i + 1$ where $\varphi_{i+1}(\bar{x}_i, \bar{y}_i) = \neg \psi(\bar{x}, \bar{y}, \bar{d})$. Then, it follows that $\mathfrak{A} \not\models \forall \bar{y} \neg \psi(\bar{c}, \bar{y}, \bar{d})$, where $\bar{c}$ are the constants used in construction step $i + 1$. This implies that $\forall \bar{y} \neg \psi(\bar{c}, \bar{y}, \bar{d})$ is inconsistent with S_i since otherwise it would be an axiom of S_ω . Consequently, $\exists \bar{x} \forall \bar{y} \neg \psi(\bar{x}, \bar{y}, \bar{d})$ is inconsistent with S_i because the constants $\bar{c}$ do not occur in S_i or $\varphi_{i+1}(\bar{x}_i, \bar{y}_i)$, and, hence, $S_i \vdash \forall \bar{x} \exists \bar{y} \psi(\bar{x}, \bar{y}, \bar{d})$. By Herbrand's theorem, the existential quantifier can be witnessed by sequences of L_ω -terms $\bar{t}_0(\bar{x}, \bar{e}), \dots, \bar{t}_k(\bar{x}, \bar{e})$, where the occurrences of L_ω -constants $\bar{e}$ are made explicit, i. e.,

$$S_i \vdash \forall \bar{x} \bigvee_{j=0}^k \psi(\bar{x}, \bar{t}_j(\bar{x}, \bar{e}), \bar{d}).$$

Replacing the constants $\bar{e}$ and $\bar{d}$ by their interpretations as parameters gives

$$\mathfrak{A} \upharpoonright L \models \forall \bar{x} \bigvee_{j=0}^k \psi(\bar{x}, \bar{t}_j(\bar{x}, \bar{e}^{\mathfrak{A}}), \bar{a}),$$

verifying Herbrand saturation. □

4.2.3 The Lower Bound

The general idea of proving Theorem 4.2.3 is to derive the consistency of the non-uniform lower bound from the truth of a uniform lower bound, stated in Lemma 4.2.14,

4 Restricted Ultrapowers

in a Herbrand-saturated model $\mathfrak{A}$ of $\forall\text{PV}$. To this end, we construct a polynomial time ultrapower over $\mathfrak{A}$ using an unbounded ultrafilter $\mathcal{U}$ and verify that for any small circuit in the ultrapower there exist hard inputs for infinitely many lengths. Note that this is sufficient because this ultrapower is itself a model of $\forall\text{PV}$ by Proposition 4.1.10. Finding such hard inputs is done by considering the circuit in the ultrapower as a function in the model $\mathfrak{A}$, the output of which is $\mathcal{U}$ -often a small circuit in $\mathfrak{A}$. Constructing from this a small uniform circuit family allows the application of the uniform lower bound stated below, yielding hard inputs for sufficiently large input lengths, which can be computed by a single function due to Herbrand saturation. It can be shown that this function interpreted as a single point in the ultrapower constitutes, up to some technical details, a hard input.

This sketch is made precise in the proof of Lemma 4.2.15. However, we first need to prove Lemma 4.2.14 providing the lower bound, which is a slight generalization of Theorem 4.2.12.

Theorem 4.2.12 (Santhanam, Williams). *For every $k \in \mathbb{N}$ there is a Boolean $g(x) \in \text{PV}$ which is not computable by PTIME -uniform size $O(n^k)$ circuit families. [13]*

The proof relies on Lemma 4.2.13, a well-known result “about a time hierarchy for deterministic time, where the lower bound holds against sublinear advice” which is proved in [13].

Lemma 4.2.13. *For every $d \in \mathbb{N}$ there exists a Boolean function $g_d \in \text{TIME}(n^{d+1})$ such that for every $h(x, y) \in \text{TIME}(n^d)$ there is a $c_h \in \mathbb{N}$ such that $\forall\text{PV}$ proves*

$$n > c_h \wedge |a| = n^{2/3} \rightarrow \exists x(|x| = n \wedge h(x, a) \neq g_d(x)).$$

Lemma 4.2.14. *Let $k \geq 3$ be a natural number, $0 < \epsilon \leq \frac{1}{3}$ a rational number, $\mathfrak{A}$ a model of $\forall\text{PV}$. Then, there exists a function $g(x) \in \text{TIME}(n^{3k})$ such that for every uniform size $n^{k+\epsilon}$ circuit family $f_a^\mathfrak{A} \in \text{PTIME}(\mathfrak{A})$ there is a Boolean function $\tilde{f}_a^\mathfrak{A} \in \text{PTIME}(\mathfrak{A})$ such that at least one of the following holds:*

$$(a) \mathfrak{A} \models \forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_{\tilde{f}_a}(|z| \cdot l^k, l))$$

$$(b) \mathfrak{A} \models \exists x(|x| = n \wedge g(x) \neq \text{eval}(f_a(1^n), x)) \text{ for sufficiently large } n \in \text{Len}(\mathfrak{A})$$

Proof. Choose g as g_{3k-1} according to Lemma 4.2.13 and let $f_a^\mathfrak{A} \in \text{PTIME}(\mathfrak{A})$ with $f(x, y) \in \text{PV}$ and $a \in A$ be any uniform size $n^{k+\epsilon}$ circuit family. We want to find a Boolean function $\tilde{f}_a^\mathfrak{A} \in \text{PTIME}(\mathfrak{A})$ such that, if case (a) fails, there is an $h(x, y) \in \text{TIME}(3k-1)$ such that for all sufficiently large $n \in \text{Len}(\mathfrak{A})$ there exists a $D \in A$ with $|D| = n^{2/3}$ such that

$$\mathfrak{A} \models \forall x(|x| = n \rightarrow h(x, D) = \text{eval}(f_a(1^n), x)). \quad (4.1)$$

This then implies case (b) since otherwise it follows for infinitely many $n \in \text{Len}(\mathfrak{A})$ that

$$\mathfrak{A} \models \forall x(|x| = n \rightarrow h(x, D) = g(x)),$$

contradicting Lemma 4.2.13.

Now, $f_a^{\mathfrak{A}}(1^n)$ codes a set of triples $\langle u, v, w \rangle$ of length at most $c \cdot |n|$ with $u, v < n^{k+\epsilon}$, $w < 3$, and some $c \in \mathbb{N}$. Then, there is a function in $\text{TIME}(n)$ that for all large enough $n \in \text{Len}(\mathfrak{A})$ takes as input a tuple $\langle n, u, v, w \rangle$ and 1^n and computes a padded variant of the tuple which has length exactly $n^{1/(2k)}$, denoted as $\langle n, u, v, w \rangle'$. Moreover, there is a function in $\text{TIME}(n)$ that computes the unpadded tuple from the padded variant and 1^n . Choose $\tilde{f}_a^{\mathfrak{A}} \in \text{PTIME}(\mathfrak{A})$ as the characteristic function of the set $\{\langle n, u, v, w \rangle' \mid n \in \text{Len}(\mathfrak{A}), \langle u, v, w \rangle \in f_a^{\mathfrak{A}}(1^n)\}$.

Assuming that (a) fails, there exist $c, d \in A$ such that for all $n \in \text{Len}(\mathfrak{A})$ with $m := n^{1/(2k)} \geq |d|$ there is a D_m of size at most $|c| \cdot n^{1/2}$ computing $\tilde{f}_a^{\mathfrak{A}}$ on inputs of length m . Furthermore, $|D_m| < n^{2/3}$ for all sufficiently large such n . Again, denote by D'_m a padded variant of length exactly $n^{2/3}$.

Choose $h(x, y) \in \text{TIME}(3k - 1)$ as the function computing the following algorithm. First, verify that $|y|^{2/3} = n := |x|$ and that y is the padded variant of some circuit D with $|D| < |y|$. If this is not the case, output 0. Otherwise, construct the circuit C coding the set

$$\{\langle u, v, w \rangle \mid u, v < n^{k+\epsilon}, w \leq 2, \text{eval}(D, \langle n, u, v, w \rangle') = 1\}$$

and output $\text{eval}(C, x)$.

This algorithm is indeed sufficiently fast because its time is dominated by the construction of the circuit C , requiring $(n^{k+\epsilon}) \cdot 3$ many evaluations of D on a padded tuple, which need time $O((|D| + |p|)^2) \leq O(n^{4/3})$ each, resulting in a running time of $O(n^{2k+2\epsilon+4/3}) \leq O(n^{3k-1})$, where the inequality follows from the choice of k and ϵ .

When giving the circuits D'_m as advice, the constructed circuit C equals $f_a^{\mathfrak{A}}(1^n)$ for all large enough $n \in \text{Len}(\mathfrak{A})$, which yields (4.1). $\square$

This lemma either provides a parameterized function satisfying the lower bound, and it only remains to be shown that this parameter can somehow be removed, as is done below in the proof of Theorem 4.2.3, or it gives us the uniform lower bound facilitating the construction of a polynomial time ultrapower in which the non-uniform lower bound is true.

Lemma 4.2.15. *Let $k \in \mathbb{N}$, $0 < \epsilon < 1$ be a rational number, $\mathfrak{A}$ a Herbrand-saturated model of $\forall\text{PV}$, and $g(x) \in \text{PV}$ a Boolean function such that for every uniform circuit family $f_a^{\mathfrak{A}} \in \text{PTIME}(\mathfrak{A})$ there exists an $N \in A$ such that for all $n \in \text{Len}(\mathfrak{A})$ with $n > N$:*

$$\mathfrak{A} \models \exists x(|x| = n \wedge g(x) \neq \text{eval}(f_a(1^n), x)).$$

Then there exists a polynomial time ultrapower over $\mathfrak{A}$ that satisfies

$$\forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_g(|z| \cdot l^k, l)).$$

Proof. Let $\Omega := \{1^n \mid n \in \text{Len}(\mathfrak{A})\}$ and $\mathcal{U}$ be any unbounded ultrafilter over Ω . We want to verify that

$$(\text{PTIME}(\mathfrak{A}) \upharpoonright \Omega) / \mathcal{U} \models \forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_g(|z| \cdot l^k, l)).$$

4 Restricted Ultrapowers

Let $\alpha = z_{a_0}^{\mathfrak{A}}, \beta = b_{a_1}^{\mathfrak{A}}$ with $z(x, y), b(x, y) \in \mathbf{PV}$ and $a_0, a_1 \in A$ be values for the variables z and b in the sentence above. Since

$$\forall x \forall y (x > p(y) \rightarrow |z(x, y)| \leq |x|^{s_0})$$

for a suitable $s_0 \in \mathbb{N}$ and polynomial bound $p(y) \in \mathbf{PV}$ is clearly true in the standard model $\mathbb{N}$ and it is a universal sentence, it is also an axiom of $\forall \mathbf{PV}$, hence, we get that $\mathfrak{A} \models |z_{a_0}(1^n)| \leq n^{s_0}$ for all sufficiently large $n \in \text{Len}(\mathfrak{A})$. Analogously, one finds some $s_1 \in \mathbb{N}$ such that $\mathfrak{A} \models |b_{a_1}(1^n)| \leq n^{s_1}$ for large enough $n \in \text{Len}(\mathfrak{A})$. For $t := \max\{\lceil \frac{s_0}{\epsilon} \rceil, s_1\}$, choose an $l(x) \in \mathbf{PV}$ with $l(1^n) = 1^{n^t}$ for all $n \in \text{Len}(\mathfrak{A})$, yielding

$$\mathfrak{A} \models |z_{a_0}(1^n)| \leq |l(1^n)|^\epsilon \quad (4.2)$$

and

$$\mathfrak{A} \models |b_{a_1}(1^n)| \leq |l(1^n)| \quad (4.3)$$

for all large enough $n \in \text{Len}(\mathfrak{A})$. Then, $\lambda := |l(x)|^{\mathfrak{A}}$ is the required witness. By (4.3)

$$(\mathbf{PTIME}(\mathfrak{A}) \restriction \Omega) / \mathcal{U} \models \lambda^{\mathcal{U}} \geq |\beta^{\mathcal{U}}|$$

because quantifier-free formulas are Łoś and $\mathcal{U}$ contains all co-bounded sets. To verify the lower bound, take any $\zeta = C_{a_2}^{\mathfrak{A}}(x)$ in $\mathbf{PTIME}(\mathfrak{A})$ with $C(x, y) \in \mathbf{PV}$ and $a_2 \in A$ such that

$$(\mathbf{PTIME}(\mathfrak{A}) \restriction \Omega) / \mathcal{U} \models \text{Circuit}(\zeta^{\mathcal{U}}, |\alpha^{\mathcal{U}}| \cdot (\lambda^{\mathcal{U}})^k) \quad (4.4)$$

and find a counter-example $\gamma \in \mathbf{PTIME}(\mathfrak{A})$, i. e.,

$$(\mathbf{PTIME}(\mathfrak{A}) \restriction \Omega) / \mathcal{U} \models |\gamma^{\mathcal{U}}| = \lambda^{\mathcal{U}} \wedge \text{eval}(\zeta^{\mathcal{U}}, \gamma^{\mathcal{U}}) \neq g(\gamma^{\mathcal{U}}).$$

By (4.4) and the fact that quantifier-free formulas are Łoś, it follows that

$$X := \langle \text{Circuit}(\zeta, |\alpha| \cdot \lambda^k) \rangle \in \mathcal{U}.$$

Since $\forall \mathbf{PV}$ supports definition by cases by Remark 4.2.7, there is a function symbol $D(x, y) \in \mathbf{PV}$ and such that

$$\mathfrak{A} \models D_{a_2}(1^n) = \begin{cases} C_{a_2}(1^{n^{1/t}}) & \text{if } \text{Circuit}(C_{a_2}(1^{n^{1/t}}), n^{k+\epsilon}) \\ 0 & \text{else} \end{cases}$$

for all $n \in \text{Len}(\mathfrak{A})$, which is a uniform circuit family of size $n^{k+\epsilon}$ in $\mathfrak{A}$ by the convention that 0 is a circuit of size 0. By assumption it follows that

$$\mathfrak{A} \models \exists x (|x| = n \wedge g(x) \neq \text{eval}(D_{a_2}(1^n), x))$$

for all sufficiently large $n \in \text{Len}(\mathfrak{A})$. In other words, there is a threshold $q \in A$ such that

$$\mathfrak{A} \models \forall y \exists x (|y| > q \rightarrow |x| = |y| \wedge g(x) \neq \text{eval}(D_{a_2}(1^{|y|}), x)),$$

which is an $\forall\exists$ -formula. Thus, applying the Herbrand saturation of $\mathfrak{A}$ yields a function symbol $w(x, y) \in \mathbf{PV}$ and a parameter $a_3 \in A$ such that

$$\mathfrak{A} \models \forall y(|y| > q \rightarrow |w_{a_3}(y)| = |y| \wedge g(w_{a_3}(y)) \neq \text{eval}(D_{a_2}(1^{|y|}), w_{a_3}(y)))$$

and in particular

$$\mathfrak{A} \models |w_{a_3}(1^n)| = n \wedge g(w_{a_3}(1^n)) \neq \text{eval}(D_{a_2}(1^n), w_{a_3}(1^n)) \quad (4.5)$$

for all large enough $n \in \text{Len}(\mathfrak{A})$. Now, take another function symbol $v(x, y) \in \mathbf{PV}$ such that $\forall \mathbf{PV}$ proves $v(x, y) = w(l(x), y)$ and thus

$$\mathfrak{A} \models v_{a_3}(1^n) = w_{a_3}(1^{n^t}),$$

and take as the counter-example

$$\gamma := v_{a_3}^{\mathfrak{A}} \in \mathbf{PTIME}(\mathfrak{A}).$$

When plugging n^t into n , it follows by (4.5) that

$$\mathfrak{A} \models |v_{a_3}(1^n)| = |l(1^n)| \wedge g(v_{a_3}(1^n)) \neq \text{eval}(D_{a_2}(l(1^n)), v_{a_3}(1^n))$$

for sufficiently large $n \in \text{Len}(\mathfrak{A})$. Since quantifier-free formulas are Łoś and $\mathcal{U}$ is unbounded, this implies for $\delta := D_{a_2}^{\mathfrak{A}}(l^{\mathfrak{A}}(x))$

$$(\mathbf{PTIME}(\mathfrak{A}) \upharpoonright \Omega) / \mathcal{U} \models |\gamma^{\mathcal{U}}| = \lambda^{\mathcal{U}} \wedge g(\gamma^{\mathcal{U}}) \neq \text{eval}(\delta^{\mathcal{U}}, \gamma^{\mathcal{U}}).$$

It remains to show that $\zeta^{\mathcal{U}} = \delta^{\mathcal{U}}$. By (4.2),

$$|z_{a_0}(1^m)| \cdot |l(1^m)|^k \leq |l(1^m)|^{k+\epsilon} = |m^t|^{k+\epsilon}$$

for all m in some co-bounded set $Y \in \mathcal{U}$. Then, for all $m \in X \cap Y$

$$\mathfrak{A} \models \text{Circuit}(C_{a_2}(1^m), (m^t)^{k+\epsilon})$$

and thus $\delta^{\mathfrak{A}}(1^m) = D_{a_2}^{\mathfrak{A}}(1^{m^t}) = C_{a_2}^{\mathfrak{A}}(1^m) = \zeta^{\mathfrak{A}}(1^m)$ by the choice of D . Finally, the ultrafilter properties of $\mathcal{U}$ imply that $X \cap Y \in \mathcal{U}$ and consequently $\delta \sim^{\mathcal{U}} \zeta$ by the definition of $\sim^{\mathcal{U}}$. $\square$

Proof of Theorem 4.2.3. Let $\mathfrak{A}$ be a Herbrand-saturated model of $\forall \mathbf{PV}$, which exists by Theorem 4.2.11, and assume without loss of generality that $k \geq 3$. Further, let $0 < \epsilon \leq \frac{1}{3}$ and $g(x) \in \mathbf{PV}$ as in Lemma 4.2.14. If the condition of Lemma 4.2.15 holds for every uniform size $n^{k+\epsilon}$ circuit family $f_a^{\mathfrak{A}} \in \mathbf{PTIME}(\mathfrak{A})$, then there is a polynomial time ultrapower over $\mathfrak{A}$ satisfying $\forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_g(|z| \cdot l^k, l))$, which in turn proves the desired consistency result because the ultrapower is a model of $\forall \mathbf{PV}$. If, on the other hand, there exists a uniform size $n^{k+\epsilon}$ family $f_a^{\mathfrak{A}} \in \mathbf{PTIME}(\mathfrak{A})$ violating the condition in Lemma 4.2.15, then Lemma 4.2.14 implies the existence of some $\hat{f}_a^{\mathfrak{A}} \in \mathbf{PTIME}(\mathfrak{A})$ such that

$$\mathfrak{A} \models \forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_{\hat{f}_a}(|z| \cdot l^k, l)). \quad (4.6)$$

4 Restricted Ultrapowers

Based on this, we take a function symbol $h(x) \in \mathbf{PV}$ such that $\forall \mathbf{PV}$ proves $h(\langle x, y \rangle) = \tilde{f}(x, y)$ and verify that

$$\mathfrak{A} \models \forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_h(|z| \cdot l^k, l)).$$

Here, the binary function symbol $\langle \cdot, \cdot \rangle \in \mathbf{PV}$ is chosen such that the function $\langle \cdot, \cdot \rangle^{\mathbb{N}}$ maps two natural numbers a, b with binary representations $a_0 \dots a_{n-1}$ and $b_0 \dots b_{m-1}$ to the natural number with binary representation $a_0 a_0 \dots a_{n-1} a_{n-1} 0 1 b_0 b_0 \dots b_{m-1} b_{m-1}$, which $\forall \mathbf{PV}$ -provably has length $2n + 2 + 2m$. It is true in $\mathbb{N}$ that there is a polynomial time computable function $p^{\mathbb{N}}$ with $p(x, y) \in \mathbf{PV}$ that takes as input a circuit C and a number y and outputs a circuit of at most the same size computing $x \mapsto C(\langle x, y \rangle)$. Hence,

$$\forall \mathbf{PV} \vdash \text{Circuit}(C, s) \rightarrow \text{Circuit}(p(C, y), s) \wedge \text{eval}(C, \langle x, y \rangle) = \text{eval}(p(C, y), x).$$

Now, if $\mathfrak{A} \not\models \forall z \forall b \exists l (l \geq |b| \wedge \text{LB}_h(|z| \cdot l^k, l))$, then there exist $c, d \in A$ such that for all $n, m \in \text{Len}(\mathfrak{A})$ with $\mathfrak{A} \models 2n + 2 + 2m \geq |d|$ there exists a circuit $C \in A$ with

$$\begin{aligned} \mathfrak{A} \models & \text{Circuit}(C, |c| \cdot (2n + 2 + 2m)^k) \\ & \wedge \forall x \forall y (|x| = n \wedge |y| = m \rightarrow \text{eval}(C, \langle x, y \rangle) = h(\langle x, y \rangle)). \end{aligned}$$

In particular, fixing $m := |a| > 0$, we choose $e_0, e \in A$ such that

$$|c| \cdot (2n + 2 + 2m)^k = |c| \cdot (2n + 2 + 2 \cdot |a|)^k \leq |e_0| \cdot n^k \cdot |a|^k \leq |e| \cdot n^k$$

for all $n \in \text{Len}(\mathfrak{A})$ with $n \geq |d|$ and get that

$$\mathfrak{A} \models \text{Circuit}(p(C, a), |e| \cdot n^k) \wedge \forall x (|x| = n \rightarrow \text{eval}(p(C, a), x) = h(\langle x, a \rangle)),$$

which contradicts (4.6) since $\mathfrak{A} \models \forall x h(\langle x, a \rangle) = \tilde{f}_a(x)$ and $n \in \text{Len}(\mathfrak{A})$ with $n > |d|$ was arbitrary, giving

$$\mathfrak{A} \models \forall l (l \geq |d| \rightarrow \neg \text{LB}_{\tilde{f}_a}(|e| \cdot l^k, l)).$$

□

5 Conclusions

The main contribution of this thesis was the development of a theory of combinatorial ultrafilters, known from set theory, in the setting of non-standard models of arithmetic, focussing on definable variants of the combinatorial properties. The most important such ultrafilters in the standard theory are Ramsey ultrafilters, P -points, and Q -points, all of which yield several equivalent characterizations. In the non-standard, definable theory, it could be shown that some relations between the analogous ultrafilters hold unconditionally, while several others could only be established under further assumptions, either about the ultrafilter or the considered structure. However, important relations remains open. The most notable question is whether d -selectivity implies the d -Ramsey property under reasonable conditions. While it is an inherently difficult problem to construct a counterexample, there is evidence supporting the conjecture that this implication is false. Such evidence is posed, first, by the fact that non-constructively showing the existence of a counterexample is indeed easy when adequately restricting the parameter set allowed for the definition of colorings and, second, the observation that mimicing the proof from standard theory requires the possibility to, intuitively speaking, definably describe certain aspects of the ultrafilter. If the conjecture holds, this would constitute a major divergence from the standard theory. The same can be said about the question whether d^n -Ramsey ultrafilters are d^{n+1} -Ramsey, which is in fact a generalization of the question regarding d -selective and d -Ramsey ultrafilters.

Beyond the relations between the different combinatorial properties, we also investigated the question which ultrafilters exist for which structures. Most remarkably, we found out that countable structures, and the more general but less intuitive small structures, admit most combinatorial ultrafilters considered in this thesis. Some existence results could also be established for uncountable models using large cardinal properties, although these again have the disadvantage of relying on axioms unprovable in ZFC. Moreover, we proved negative results showing the existence of structures without d -Ramsey ultrafilters.

Finally, a slight generalization of the lower bound consistency from [1] has been proved. This proof primarily served as a demonstration of how restricted ultrapowers can be utilized for proving consistency results for the theory $\forall\text{PV}$ from bounded arithmetic and did not involve an application of combinatorial ultrafilter properties.

This leads to the possible directions of future research. On the one hand, gaining a better understanding of the theory of combinatorial ultrafilters in the non-standard, definable setting is interesting in its own right, just like finding more general and extensive conditions for their existence over uncountable structures. On the other hand, this whole theory is essentially motivated by the search for more powerful ultrapower constructions. Thus, the main goal of research building upon non-standard combinatorial

5 *Conclusions*

ultrafilters should be to find such an application as this could, in the best case, lead to a new technique of model-theoretic consistency proofs.

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